



asi

Reference Manual

Version 1.29

05/06/2025

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1 Overview

Changelog:

1.29

- **Changed**
 - asiStatus from 1.5 to 1.6
 - status from 1.23 to 1.24
 - srt from 1.5 to 1.6
 - input from 1.33 to 1.35
 - output from 1.35 to 1.36

1.28

- **Changed**
 - status from 1.22 to 1.23
 - asiStatus from 1.4 to 1.5

1.27

- **Changed**
 - input from 1.30 to 1.33;
 - output from 1.33 to 1.35;
 - srt from 1.4 to 1.5;
 - srtValidate from 1.0 to 1.1;
 - bissDescrambling from 1.5 to 1.6;
 - bissTypes from 1.2 to 1.3;
 - bissVersion1 from 1.2 to 1.3;
 - bissVersion2 from 1.5 to 1.6;

1.26

- **Added**
 - srtCardSettings 1.0
- **Changed**
 - srt from 1.3 to 1.4
 - input from 1.29 to 1.30
 - output from 1.32 to 1.33

1.25

- **Changed**
 - asiStatus from 1.3 to 1.4
 - status from 1.21 to 1.22
 - srt from 1.2 to 1.3
 - input from 1.28 to 1.29

- output from 1.31 to 1.32

1.24

- **Changed**
 - input from 1.27 to 1.28

1.23

- **Changed**
 - asiStatus from 1.2 to 1.3
 - status from 1.20 to 1.21

1.22

- **Changed**
 - input from 1.26 to 1.27

1.21

- **Changed**
 - input from 1.25 to 1.26

1.20

- **Changed**
 - tsstatus from 1.3 to 1.5
 - status from 1.19 to 1.20
 - componentMapping from 1.2 to 1.3
 - asiStatus from 1.1 to 1.2
 - output from 1.30 to 1.31
 - outputMapping from 1.3 to 1.4
 - genericDescriptor from 1.1 to 1.2
 - mappingTemplates from 1.1 to 1.2

1.19

- **Added**
 - asiBissSettings version 1.1
- **Removed**
 - bissData

1.18

- **Added**
 - asiTools version 1.0
- **Changed**
 - input from 1.23 to 1.25
 - output from 1.28 to 1.30
 - outputMapping from 1.2 to 1.3

- srt from 1.1 to 1.2
- bissScrambling from 1.4 to 1.5
- bissDescrambling from 1.4 to 1.5
- bissVersion2 from 1.4 to 1.5

1.17

- **Changed**

- status from 1.18 to 1.19
- input from 1.22 to 1.23

1.16

- **Added**

- asiTriggers version 1.0
- triggers version 1.0
- baseUuidNamespace version 1.0
- asiStatus version 1.0

- **Changed**

- output from 1.26 to 1.27
- input from 1.20 to 1.22
- status from 1.17 to 1.18
- asiModule from 1.10 to 1.11

1.15

- **Changed**

- output from 1.25 to 1.26
- input from 1.19 to 1.20
- status from 1.16 to 1.17
- asiModule from 1.9 to 1.10

1.14

- **Added**

- streamtypes version 1.0
- mappingTemplates 1.1;

- **Changed**

- component from 1.1 to 1.4;

- **Removed**

- ipinterface 1.4
- physicalports 1.4

- **Changed**

- output from 1.24 to 1.25;

- input from 1.18 to 1.19;
- output from 1.24 to 1.25;
- tsstatus from 1.1 to 1.3;
- status from 1.15 to 1.16;
- asiModule from 1.8 to 1.9;
- outputMapping from 1.0 to 1.1;
- genericDescriptor from 1.0 to 1.1;
- srt from 1.0 to 1.1
- zixi from 1.0 to 1.1

1.13

- **Added**

- bissCaSettings 1.0;

- **Changed**

- output from 1.22 to 1.24;
- input from 1.17 to 1.18;
- bissVersion2 from 1.3 to 1.4;
- bissScrambling from 1.3 to 1.4;
- bissDescrambling from 1.3 to 1.4;
- asiModule to 1.8

1.12

- **Added**

- bissData 1.2;
- bissHelpers 1.0;
- bissScrambling 1.3;
- bissDescrambling 1.3;
- bissTypes 1.2;
- bissSettings 0.0;
- bissValidate 1.0;
- bissVersion1 1.2;
- bissVersion2 1.3;

- **Changed**

- input from 1.14 to 1.17
- status from 1.12 to 1.14
- output from 1.19 to 1.22
- asiModule from 1.5 to 1.7

2 asiBissSettings (1.1)

3 asiModule (1.11)

3.1 Overview

Changelog:

1.11

- **Added**
 - Added protocol GetAsiInputStatusProtocol
 - Added protocol GetAsiOutputStatusProtocol
- **Changed**
 - Split asiModule into asiModule and asiStatus.
 - The field 'AsiBitrates' replaced the fields 'totalBitsPerSecond' and 'effBitsPerSecond' in AsiStatus.
 - The field 'AsiBitrates' replaced the fields 'totalBitsPerSecond' and 'effBitsPerSecond' in AsiOutputStatus.

1.10

- **Changed**
 - The imported version of status was bumped to 1.17

1.9

- **Changed**
 - The imported version of status was bumped to 1.16
 - The imported version of tsstatus was bumped to 1.3

1.8

- **Changed**
 - The imported version of status was bumped to 1.15

1.7

- **Changed**
 - The imported version of status was bumped to 1.14

1.6

- **Changed**
 - The imported version of status was bumped to 1.13.
 - The imported version of tsstatus was bumped to 1.1.

1.5

- **Changed**
 - The imported version of status was bumped to 1.12.

3.2 Command Reference

3.2.1 GetAsiPhysicalPorts

- message **GetAsiPhysicalPorts.Request**
- message **GetAsiPhysicalPorts.Response**

3.2.2 SetAsiPhysicalPorts

- message **SetAsiPhysicalPorts.Request**
- message **SetAsiPhysicalPorts.Response**

3.2.3 ClearCounters

- message **ClearCounters.Request**
- message **ClearCounters.Response**

3.3 Type Reference

3.3.1 AsiConfigByteMode

Config enumeration. The ASI byte mode affects the maximum bit rate. Spread mode limits the max bit rate to 72Mbit/s, whereas max bit rate for burst depends on packet size.

enum

ASI_BURST	
ASI_SPREAD	

3.3.2 AsiConfigPacketSize

Config enumeration. The ASI packet size affects the maximum bit rate when in ASI burst mode.

enum

ASI_PACKET_SIZE_188	
ASI_PACKET_SIZE_204	

3.3.3 AsiPhysicalPort

AsiPhysicalPort contains the configuration for the physical BNC port on the card.

struct

enabled	bool Whether or not the port should be enabled in the FPGA.
portNumber	int The port index (0-7) of the port.
label	string A label string to associate with this port.
direction	PortDirection The configured direction/mode of the port.

3.3.4 AsiTstGenerator

Type to hold the test signal generator information for ASI Input cards.

struct

enabled	bool When true, the input stream will be replaced with NULL packets of constant bitrate.
autoMode	bool When true, test generator will only operate upon signal lost.
cbr	int Constant bitrate of the generated signal, in units of (kbps). Allowed range is 0 - 213 mbps.

3.3.5 AsiTrafPolicing

Type to hold the traffic policing information for ASI Input cards.

struct

enabled	bool When true, the input stream will be taken down if the input bitrate exceeds maxBitrate.
maxBitrate	int Max allowed bitrate for traffic policing, in units of (kbps). Allowed range is 0 - 213 mbps.

3.3.6 AsiTsrFlywheel

Type to hold the TSR flywheel information for ASI Output cards.

struct

enabled	bool When true, bitrate of the output stream is going to be preserved with NULL packets, in case the input source bitrate is not stable.
continuous	bool When true, the output stream bitrate will be maintained continuously.
duration	int Shows how long the output stream bitrate should be maintained, in case continuous is set to false. In units of milliseconds, allowed range is 0 - 30 seconds.
maxAllowedPktDrops	int Max number of allowed packet drops before TSR Flywheel kicks in, in range [1, 15]

3.3.7 ClearCounters.Request

ClearCountersRequest

3.3.8 ClearCounters.Response

ClearCountersResponse

3.3.9 ClearCountersRequest

The Clear Counters request has no parameters.

empty **struct**

3.3.10 ClearCountersResponse

The Clear Counters response has no parameters.

empty **struct**

3.3.11 GetAsiPhysicalPorts.Request

The ASI Physical Ports GET request has no parameters.

empty **struct**

3.3.12 GetAsiPhysicalPorts.Response

The ASI Physical Ports GET response returns a map from each port to its corresponding AsiPhysicalPort object.

struct

data

map from **UUID** to **AsiPhysicalPort**

A map from port UUID to the AsiPhysicalPort object

3.3.13 InputAsi

Input object for ASI, which works as a proxy for either seamless mode or single input.

variant

single

InputAsiSettings

3.3.14 InputAsiSettings

Settings object for ASI Input.

struct

id

UUID

ID of this ASI Input node.

portId

UUID

ID of the physical port which is the source of this Input.

portNumber

int

testGenerator

AsiTestGenerator

trafficPolicing

AsiTrafficPolicing

int

bufferDelay

Buffer delay to align with other sources to tune the total latency, in units of (ms). Allowed range is 0-1500ms. Default 0ms.

3.3.15 InputAsiTransportSettings

Top level ASI input configuration type, which is used in the transportSettings level of the generic Input objects.

struct

input

InputAsi

3.3.16 OutputAsi

ASI output variant that selects either cloned or single output mode.

variant

single

OutputAsiSettings

Configuration for the only port used by this output.

3.3.17 OutputAsiSettings

Output settings for ASI outputs.

struct

id

UUID

ID of this ASI Output node.

portId

UUID

ID of the physical port which is the destination of this Output.

portNumber

int

byteMode

AsiConfigByteMode

Byte mode can be either burst or spread.

packetSize

AsiConfigPacketSize

Packet size can be either 188 or 204 bytes.

tsrFlywheel

AsiTsrFlywheel

3.3.18 OutputAsiTransportSettings

Transport settings for an ASI output.

struct

output

OutputAsi

Output settings for ASI, which can contain either single or cloned port.

3.3.19 PortDirection

ASI physical port directions can be configured to either input, output, passthrough (monitoring, input only) or cloned (output only).

enum

INPUT

OUTPUT

PASSTHROUGH

CLONED

3.3.20 SetAsiPhysicalPorts.Request

The ASI Physical Ports SET request requires a map from each port to its corresponding AsiPhysicalPort object.

struct

data

map from **UUID** to **AsiPhysicalPort**

A map from port UUID to the AsiPhysicalPort object

3.3.21 SetAsiPhysicalPorts.Response

The ASI Physical Ports SET response has no parameters.

empty **struct**

4 asiStatus (1.6)

4.1 Overview

Changelog:

1.6

- Changed
 - status from 1.23 to 1.24

1.5

- Changed
 - status from 1.22 to 1.23

1.4

- Changed
 - status from 1.21 to 1.22

1.3

- Changed
 - status from 1.20 to 1.21

1.2

- Changed
 - tsstatus from 1.3 to 1.5
 - status from 1.19 to 1.20

4.2 Command Reference

4.2.1 GetAsiInputPidStatus

Get Pid status for a set of inputs.

- message **GetAsiInputPidStatus.Request**
- message **GetAsiInputPidStatus.Response**

4.2.2 GetAsiStatus

- message **GetAsiStatus.Request**
- message **GetAsiStatus.Response**

4.2.3 GetAsiOutputStatus

- message **GetAsiOutputStatus.Request**
- message **GetAsiOutputStatus.Response**

4.3 Type Reference

4.3.1 AsiBitrates

struct

total	AsiTotalBitrates
effective	AsiEffectiveBitrates

4.3.2 AsiEffectiveBitrates

struct

effectiveFlowBitrate	int
effectiveFlowBits	bigint

4.3.3 AsiOutputStatus

struct

outputNumber	int Which output port the status belongs to (zero-indexed, 0-7).
carrierDetected	bool False = No carrier detected, True = Carrier detected.
bitrates	AsiBitrates Total and effective bitrates of the output in bits and bits per second
outputSources	list of status.OutputSourceStatus

4.3.4 AsiStatus

The ASI status type contains ASI input status values from the fwcom131 status block.

struct

carrierDetected	bool Whether or not the cable is connected and has a carrier signal.
asiInputNumber	int Which input port the status belongs to (zero-indexed, 0-7).
asiSignalDetected	bool False = No signal detected, True = Signal detected by cable equalizer.
asi10BLink	bool False = Unable to frame the serial signal, True = The deserializer is locked.
asiSignalLostCount	int The number of times the signal was lost (every negative edge of asiSignalDetected).
bitrates	AsiBitrates Total and effective bitrates of the output in bits and bits per second
byteMode	AsiStatusByteMode Can be either ASI Spread mode, or ASI Burst mode.

packetSize	AsiStatusPacketSize Can be either 188 or 204.
syncByteErrorCount	int The number of times sync byte is not equal to 0x47 when transport stream is synced.
teiCount	int
bitErrorCount	int The number of code/disparity errors (One 8-bit data value) reported by the decoder.
tsSyncLostCount	int The number of times transport stream sync was lost.
tsPktLostNoBuffer	int Number of transport stream packets lost due to no buffer.

4.3.5 AsiStatusByteMode

Status enumeration. The ASI byte mode affects the maximum bit rate. Spread mode limits the max bit rate to 72Mbit/s, whereas max bit rate for burst depends on packet size.

enum

```
UNKNOWN
ASI_BURST
ASI_SPREAD
```

4.3.6 AsiStatusPacketSize

Status enumeration. The ASI packet size affects the maximum bit rate when in ASI burst mode.

enum

```
UNKNOWN
ASI_PACKET_SIZE_188
ASI_PACKET_SIZE_204
ASI_PACKET_SIZE_204_AND_188
```

4.3.7 AsiTotalBitrates

struct

```
totalFlowBitrate    int
totalFlowBits       bigint
```

4.3.8 GetAsiInputPidStatus.Request

PidStatusRequest

4.3.9 GetAsiInputPidStatus.Response

PidStatusResponse

4.3.10 GetAsiOutputStatus.Request

GetAsiOutputStatusRequest

4.3.11 GetAsiOutputStatus.Response

GetAsiOutputStatusResponse

4.3.12 GetAsiOutputStatusRequest

The ASI Output Status GET request has no parameters.

empty **struct**

4.3.13 GetAsiOutputStatusResponse

The ASI Output Status response returns a map from each port to its corresponding AsiOutputStatus object.

struct

data

map from **UUID** to **AsiOutputStatus**

A map from port UUID to the AsiOutputStatus object corresponding to that port.

4.3.14 GetAsiStatus.Request

GetAsiStatusRequest

4.3.15 GetAsiStatus.Response

GetAsiStatusResponse

4.3.16 GetAsiStatusRequest

The ASI Status GET request has no parameters.

empty **struct**

4.3.17 GetAsiStatusResponse

The ASI Status GET response returns a map from each port to its corresponding AsiStatus object.

struct

data

map from **UUID** to **AsiStatus**

A map from port UUID to the AsiStatus object corresponding to that port.

4.3.18 PidStatusRequest

struct

inputs

list of UUID

4.3.19 PidStatusResponse

struct

data

map from UUID to tsstatus.TransportStreamStatus

5 asiTools (1.0)

6 asiTriggers (1.0)

7 asiValidate (1.0)

7.1 Type Reference

7.1.1 FloatRange

Float Range

struct

min	float Minimum value of the range
max	float Maximum value of the range

7.1.2 IntRange

Integer Range

struct

min	int Minimum value of the range
max	int Maximum value of the range

7.1.3 Result

Validation result

struct

result	bool True if config is valid and false otherwise.
msg	optional string An error message, which may be given if the result is invalid.

8 baseUuidNamespace (1.0)

9 bissCaSettings (1.0)

9.1 Type Reference

9.1.1 BissCaEcmInterval

BISS CA ECM Repetition Interval

Predefined set of possible ECM repetition interval values, in milliseconds.

Values lower than 100 ms are outside of the BISS CA specification. These values may still be used if the parameter `extendedOptions` is set to true in the `BissCaSettings` struct.

enum

I6	5.7 ms (Value outside BISS CA spec)
I15	14.7 ms (Value outside BISS CA spec)
I50	50 ms (Value outside BISS CA spec)
I100	100 ms
I125	125 ms
I150	150 ms
I175	175 ms
I200	200 ms
I250	250 ms
I300	300 ms

9.1.2 BissCaEntitlementFlags

BISS CA Entitlement Flags

The entitlement flags are parameters for how the entitled receivers shall behave. These values are mandatory part of the stream, and is inserted into the session data, which is the encrypted part of the EMM table.

struct

<code>preventDescrambledForward</code>	bool Prevent descrambling forward
<code>preventDecodedForward</code>	bool Prevent decode forward
<code>insertWatermark</code>	bool Insert watermark

9.1.3 BissCaSettings

BISS CA Specific Settings

struct

<code>cryptoPeriod</code>	int Crypto period (in seconds).
<code>emmBitrate</code>	int EMM bitrate (kbps).
	BissCaEcmInterval

ecmInterval	ECM repetition interval. Note that changes to the repetition interval will not take effect until the next crypto period (the next time the ECM content is updated).
emmPreferredPid	int EMM preferred PID value. If the PID value is not available, the first larger PID value available will be used instead. If no larger PID value is available, then the first smaller available value will be used.
ecmPreferredPid	int ECM preferred PID value. If the PID value is not available, the first larger PID value available will be used instead. If no larger PID value is available, then the first smaller available value will be used.
entitlementFlags	BissCaEntitlementFlags BISS CA entitlement flags. Controls how the entitled receivers shall behave.
extendedOptions	bool Allow for extended options. This means allowing violation of the BISS CA specification. If set to true, then the following requirements of the BISS CA specification are <i>not</i> enforced: - EMM repetition interval must be at least 200 ms - ECM repetition interval must be at least 100 ms Note that the requirement that EMM interval must be larger than the ECM interval is still enforced.

10 bissDescrambling (1.6)

10.1 Type Reference

10.1.1 DescramblerSettings

- Descrambler Settings
-
- Settings defining how to descramble the transport stream.
- By default, there is a single descrambling settings parameter, where these
- settings are applied to all services in the transport stream (TS).
-
-
-

struct

enabled	bool Descrambling on/off. •
---------	--

method	DescramblingMethod Descrambling settings.
--------	---

10.1.2 DescramblingMethod

DescramblingMethod

variant

fixedKeyRaw	bissTypes.FixedKeyRawDescrambling Fixed key decryption.
-------------	---

biss1	bissVersion1.Descrambling BISS version 1 decryption.
-------	--

biss2	bissVersion2.Descrambling BISS version 2 decryption.
-------	--

11 bissHelpers (1.0)

12 bissScrambling (1.6)

12.1 Type Reference

12.1.1 ScramblerSettings

- Scrambler Settings
-
- Settings defining how to scramble the transport stream.
- By default, there is a single scrambling settings parameter, where these
- settings are applied to all services in the transport stream (TS).
-
- Optionally, however, one or more extra settings may be configured, where the
- different settings are applied to different services in the TS. The different
- settings are referenced by integer indecies. These indecies are
- cross-references to the transport stream configuration. For example, a single
- service may refer to one of these indecies, meaning that the service will be
- scrambled with the corresponding scrambling settings.
-
- Requirements:
 - – All extra scrambling setting *must* contain the same scrambling method as
- the main settings.
-
- If these requirements are not met, the configuration is not valid.
-
-

struct

enabled	bool Scrambling on/off •
main	ScramblingMethod The main (default) scrambling settings •
extra	map from int to ScramblingMethod Additional scrambling settings, each with its own index.

12.1.2 ScramblingMethod

ScramblingMethod

variant

fixedKeyRaw	bissTypes.FixedKeyRawScrambling Fixed key encryption
biss1	bissVersion1.Scrambling BISS version 1 encryption
biss2	bissVersion2.Scrambling BISS version 2 encryption

13 bissSettings (1.0)

13.1 Type Reference

13.1.1 AllowedExtraSettings

Allowed Num of Extra Scrambling Settings

enum

EXTRA_15_ECMS	15 extra settings
EXTRA_7_ECMS	7 extra settings
EXTRA_3_ECMS	3 extra settings

13.1.2 AllowedMode

Allowed (De)Scrambling modes

enum

ALL	CSA and AES algorithms allowed
AES_ONLY	Only AES allowed
CSA_ONLY	Only CSA allowed

13.1.3 BissModuleSettings

BISS Module Setup for a given product (not configurable)

struct

allowedMode	AllowedMode Algorithms supported by product
numExtraPerOutput	AllowedExtraSettings Number of extra settings allowed per output
maxNumOutputs	int Maximum number of outputs
totalEmmBitrateLimit	optional int Set an upper limit for the total EMM playout bitrate (sum of the bitrate all EMMs). Given in units of kbps.

14 bissTypes (1.3)

14.1 Type Reference

14.1.1 Algorithm

De/Scrambling Algorithm

The DVB-CSA algorithm uses a 64-bit key, while all the AES algorithms use a 128-bit key.

enum

DVB_CSA	DVB-CSA
AES_ATIS_IDSA	AES ATIS IDSA
AES_ECBT_ECB1	AES-ECBT/AES-ECB1 (Alticast)
AES_ECBL_ECB2	AES-ECBL/AES-ECB2 (Latens)
AES_CBCT_CBC1	AES CBCT CBC1
AES_DVB_CISSA	AES DVB-CISSA
AES_IRDETO_CBCT	AES CBCT Irdeto
AES_CBCT_IV	AES CBCT w/specified Initialization Vector

14.1.2 FixedKeyRawDescrambling

Descrambling Method: Fixed Key Raw

struct

algorithm	Algorithm Scrambling algorithm
key	string Scrambling key
iv	optional string Scrambling Initialization vector (only for AES_CBCT_IV)
serviceMappings	list of FixedKeyServiceMapping A list of service mappings.

14.1.3 FixedKeyRawScrambling

Scrambling Method: Fixed Key Raw

struct

algorithm	Algorithm Scrambling algorithm
key	string Scrambling key
iv	optional string Scrambling Initialization vector (only for AES_CBCT_IV)

14.1.4 FixedKeyServiceMapping

Fixed key descrambling for a list of services

struct

key	string Scrambling key
iv	optional string Scrambling Initialization vector (only for AES_CBCT_IV)
serviceIds	list of int A list of services that the key applies to.

15 bissValidate (1.0)

15.1 Type Reference

15.1.1 FloatRange

Float Range

struct

min	float Minimum value of the range
max	float Maximum value of the range

15.1.2 IntRange

Integer Range

struct

min	int Minimum value of the range
max	int Maximum value of the range

15.1.3 Result

Validation result

struct

result	bool True if config is valid and false otherwise.
msg	optional string An error message, which may be given if the result is invalid.

16 bissVersion1 (1.3)

16.1 Type Reference

16.1.1 CsaKeyExpansion

Method of CSA Key Expansion from 48 to 64 bit

enum

CHECKSUM	Session Word is expanded using checksum.
ZERO	Session Word is expanded using zeros.

16.1.2 Descrambling

BISS version 1 Descrambling

variant

mode1	Mode1Descrambling BISS version 1, Mode 1.
modeE	ModeEDescrambling BISS version 1, Mode E.

16.1.3 Mode1Descrambling

BISS version 1 Descrambling, Mode 1

struct

expansion	CsaKeyExpansion Expansion method.
sessionWord	string Session Word, 48-bit hex string.
serviceMappings	list of Mode1ServiceMapping A list of service mappings.

16.1.4 Mode1Scrambling

BISS version 1 Scrambling, Mode 1

struct

expansion	CsaKeyExpansion Expansion method.
sessionWord	string Session Word, 48-bit hex string.

16.1.5 Mode1ServiceMapping

BISS version 1 Descrambling, Mode 1 for a list of services

struct

sessionWord	string Session Word, 48-bit hex string.
serviceIds	list of int A list of services that the session word applies to.

16.1.6 ModeEDescrambling

BISS version 1 Descrambling, Mode E

struct

activeId	UUID Active session ID.
expansion	CsaKeyExpansion Expansion method.
encSessionWord	UUID Encrypted Session Word, 128-bit. Reference to BissData UUID.
serviceMappings	list of ModeEServiceMapping A list of service mappings.

16.1.7 ModeEScrambling

BISS version 1 Scrambling, Mode E

struct

expansion	CsaKeyExpansion Expansion method.
sessionWord	UUID Session Word, 128-bit. Reference to BissData UUID.

16.1.8 ModeEServiceMapping

BISS version 1 Descrambling, Mode E for a list of services

struct

activeId	UUID Active session ID. Reference to BissData UUID.
encSessionWord	UUID Encrypted Session Word, 128-bit. Reference to BissData UUID.
serviceIds	list of int A list of services that the session ID and encrypted session word applies to.

16.1.9 Scrambling

BISS version 1 Scrambling

variant

mode1	Mode1Scrambling BISS version 1, Mode 1.
-------	---

modeE	ModeEScrambling BISS version 1, Mode E.
-------	---

17 bissVersion2 (1.6)

17.1 Type Reference

17.1.1 Descrambling

BISS version 2 Decrambling

variant

mode1	ModelDescrambling BISS version 2, Mode 1.
modeE	ModeEDescrambling BISS version 2, Mode E.
modeCA	ModeCADescrambling BISS version 2, Mode CA.

17.1.2 ModelDescrambling

BISS version 2 Descrambling, Mode 1

struct

sessionWord	string Session Word, 128-bit hex string.
serviceMappings	list of ModelServiceMapping A list of service mappings.

17.1.3 ModelScrambling

BISS version 2 Scrambling, Mode 1

struct

sessionWord	string Session Word, 128-bit hex string.
-------------	--

17.1.4 ModelServiceMapping

BISS version 2 Descrambling, Mode 1 for a list of services

struct

sessionWord	string Session Word, 128-bit hex string.
serviceIds	list of int A list of services that the session word applies to.

17.1.5 ModeCADescrambling

BISS version 2 Decrambling, Mode CA

struct

key

optional UUID

Private RSA key. If UUID value is given, this references a Biss-Data UUID. If optional is empty, the buried RSA private key is used.

17.1.6 ModeCAScrambling

BISS version 2 Scrambling, Mode CA

struct

receivers

UUID

Public RSA keys of receivers. Reference to BissData CSV File UUID or BissDataGroup UUID.

esId

string

Entitlement session ID (16-bit hex string).

settings

bissCaSettings.BissCaSettings

ModeCa settings.

17.1.7 ModeEDescrambling

BISS version 2 Descrambling, Mode E

struct

activeId

UUID

Active session ID. Reference to BissData UUID.

encSessionWord

UUID

Encrypted Session Word, 128-bit. Reference to BissData UUID.

serviceMappings

list of ModeEServiceMapping

A list of service mappings.

17.1.8 ModeEScrambling

BISS version 2 Scrambling, Mode E

struct

sessionWord

UUID

Session Word, 128-bit. Reference to BissData UUID.

17.1.9 ModeEServiceMapping

BISS version 2 Descrambling, Mode E for a list of services

struct

activeId

UUID

Active session ID. Reference to BissData UUID.

encSessionWord

UUID

Encrypted Session Word, 128-bit. Reference to BissData UUID.

serviceIds

list of int

A list of services that the session ID and encrypted session word applies to.

17.1.10 Scrambling

BISS version 2 Scrambling

variant

mode1

Mode1Scrambling

BISS version 2, Mode 1.

modeE

ModeEScrambling

BISS version 2, Mode E.

modeCA

ModeCAscrambling

BISS version 2, Mode CA.

18 component (1.5)

18.1 Overview

Changelog:

1.5

- Added
 - SCTE27_SUB to ComponentType enum and respective mappings.

18.2 Type Reference

18.2.1 AnyComponent

empty **struct**

18.2.2 ComponentGroup

ComponentGroup: Grouping of ComponentType, used for filtering

enum

UNDEFINED
AUDIO
VIDEO
SUBTITLE
DATA

18.2.3 ComponentLanguage

string

18.2.4 ComponentSelector

Component selector.

@param pid	Selects a component using a specific (source) PID.
@param language	Selects a component by a three character language string.
@param type	Selects a component by its type.
@param group	Selects a component by group membership.
@param any	Selects any component.

variant

pid	Pid
language	ComponentLanguage
componentType	ComponentType
componentGroup	ComponentGroup
any	AnyComponent

18.2.5 ComponentType

A ComponentType is typically identified by a stream type (see ISO 13818-1 Table 2-29), often distinguished with another header field such as a descriptor tag.

enum

MPEG_VIDEO
MPEG_AUDIO
TTX
DVBSUB
ECM
PCR
PRIVATE
AC_3
H264
VBI
AAC
VC_1
SCS_EMM
DPI_CUE
EAC3
AIT
HBBTV_CAROUSEL
DATA_CAROUSEL
H265
S302M
JPEG2000
S2038
DVBTTMLSUB
JPEG_XS_VIDEO
SCTE27_SUB

18.2.6 Pid

int

18.2.7 SubtitleFormat

enum

DVB
DVB_HARD_OF_HEARING
EBU
EBU_HARD_OF_HEARING

19 componentMapping (1.3)

19.1 Overview

Changelog:

1.3

- **Changed**
- component bumped from 1.4 to 1.5

1.2

- **Changed**
- Moved the StreamType definitions to dedicated IDCL module, streamtype
- Now using the ComponentType definition found in IDCL module component, as the one found in componentMapping version 1.1 was redundant.

19.2 Type Reference

19.2.1 ComponentAction

enum

STOP

PASSTHROUGH

REMAP

19.2.2 ComponentGeneration

A pmt entry generation rule. Usually used to indicate that some external PID will Appear at the specified PID.

struct

streamType

streamtype.StreamType

Streamtype as specified in the standards.

pid

int

PID number to signal the component on.

19.2.3 ComponentLayoutMap

A collection of a priority list of component mapping rules, and one default action to handle unmatched components.

struct

defaultAction

ComponentMapDefaultAction

Default action to perform on any unmatched component

layout

list of **ComponentMapping**

Priority list of mapping rules - each rule matches exactly one source component

19.2.4 ComponentMapDefaultAction

enum

PASSTHROUGH
STOP

19.2.5 ComponentMapping

The ComponentMapping object describes a rule that will match one - and only one - incoming component, and perform an action on it. I.e, the selection of type MPEG_AUDIO with language "nor" with an action set to "STOP", will stop one "instance" of mpeg audio with the norwegian language. Subsequent components matching the same filter will fall through to the next lower priority rule matching them.

There are some extra considerations in regards to pids containing PCR. It is not allowed to STOP the PCR content. If such a rule is applied, the payload of the component will be stopped, but the PCR content will still be mapped out.

It is possible to remap only the PCR content. To do this, componentType must be specified as PCR. Any other rule matching the same pid will remap all the content, not just the PCR.

struct

componentType	optional component.ComponentType Component type selector.
language	optional string Language selector.
pid	optional int Pid selector.
action	ComponentAction What to do with the matched component.
targetPid	optional int Optional. Must be specified when action is REMAP.

20 firewallTypes (1.0)

20.1 Type Reference

20.1.1 IPAddress

string

20.1.2 PaginatedRequest

Instructs the server to return only a subset of the data found on the server

@param pageSize number of entries returned per request

@param pageNumber what page of the set to returned, determined by the slice [pageSize*pageNumber

struct

pageSize	int
pageNumber	int

20.1.3 PaginationInfo

struct

totalEntries	int
--------------	-----

the total number of entries on the server, before matching

20.1.4 SocketAddress

Tuple with IP address and port

struct

address	IPAddress
port	int

IP address, either IPv4 or IPv6

20.1.5 UUID

string

21 gatewayValidate (1.0)

21.1 Type Reference

21.1.1 BigIntRange

struct

min	bigint
max	bigint

21.1.2 FloatRange

Float Range

struct

min	float Minimum value of the range
max	float Maximum value of the range

21.1.3 IntRange

Integer Range

struct

min	int Minimum value of the range
max	int Maximum value of the range

21.1.4 Result

Validation result

struct

result	bool True if config is valid and false otherwise.
msg	optional string An error message, which may be given if the result is invalid.

22 genericDescriptor (1.2)

22.1 Overview

Changelog:

1.2

- **Changed**
 - The imported component version was bumped to 1.5.

1.1

- **Changed**
 - The imported component version was bumped to 1.4.

22.2 Type Reference

22.2.1 DescriptorAction

Dictate what action is to be performed on the provided descriptor.

enum

ADD
REMOVE
CHANGE

22.2.2 GenericPmtDescriptorEntry

Defines a generic descriptor entry with a given action, tag and optional data.

@param action	What action is to be performed with this descriptor entry.
@param tag	The Descriptor tag.
@param component	Optional component (inner loop) this descriptor should apply to.
@param data	Optional data to add to/replace the descriptor.

struct

action	DescriptorAction
tag	int
component	optional component.ComponentSelector
payload	optional string

23 input (1.35)

23.1 Overview

Changelog:

1.35

- **Changed**
- The imported `srt` version was bumped to 1.6.

1.34

- **Changed**
- Removed is100G parameter from getDefaultInputBufferSize function.
- Changed seamless buffer value from 100 to 20 for 100G.
- Changed default dejitter buffer value from 100 to 40 for 100G.

1.33

- **Changed**
 - The imported bissDescrambling version was bumped to 1.6.

1.32

- **Changed**
 - The imported srt version was bumped to 1.5.

1.31

- **Changed**
- Added new is100G parameter to getDefaultInputBufferSize function.
- Added new getDefaultDejitterBufferSize function.
- 100G min, max values for Traffic Policer

1.30

- **Changed**
- The imported `srt` version was bumped to 1.4.

1.29

- **Changed**
- The imported `srt` version was bumped to 1.3.

1.28

- **Changed**
- Added optimizeLatency to InputUdpSeamless

1.27

- **Changed**
- Make descrambling optional under AdditionalFeatures
- Add triggers to AdditionalFeatures

1.26

- ****Changed****
 - The field 'profile' in 'DejitterRtp' is changed from an int to new enum type 'RtpDejitterPro

1.25

- ****Changed****
 - The imported bissDescrambling version was bumped to 1.5
 - Function getMaxBitrateRange is changed to getIpTrafficPolicingBitrateValidRange with new max
- ****Added****
 - Function validateIpTrafficPolicingBitrate [15.11.22]

1.24

- ****Added****
 - Validation for the field 'inputBufferSize' in 'InputUdpTransportSettings'.
 - Validation for the field 'bufferSize' in 'Dejitter'.
 - Validation for the field 'port.destination' in 'InputUdpSettings'.
- ****Changed****
 - The imported srt version was bumped to 1.2
 - The field 'enabled' in 'IpTrafficPolicing' is now checked before the field 'maxBitrate' is v

1.23

- **Added**
 - The field 'rptExcessiveLossThreshold' was added to 'InputUdpTransportSettings'.
 - The field 'alarmWhenScrambled' was added to 'InputUdpTransportSettings'.

1.22

- **Added**
 - The field 'rtp' was added to 'InputUdpSettings'.

1.21

- **Changed**
 - The imported asiModule version was bumped to 1.11

1.20

- **Changed**
 - The imported asiModule version was bumped to 1.10

1.19

- **Changed**
 - The imported asiModule version was bumped to 1.9.
 - The imported srt version was bumped to 1.1.
 - The imported zixi version was bumped to 1.1.

1.18

- **Changed**
 - The imported bissDescrambling version was bumped to 1.4.
 - The imported asiModule version was bumped to 1.8

1.17

- **Added**

- The optional field `additionalFeatures` was added to `Input`.

The `AdditionalInputFeatures` type includes the field `descrambling`, which is used to configure descrambling on the input.

- **Changed**

- The imported `asiModule` version was bumped to 1.7.

- **Changed**

- The imported `asiModule` version was bumped to 1.7.

1.16

- **Changed**

- The imported `asiModule` version was bumped to 1.6.

1.15

- **Removed**

- The field `cbrBitRate` was removed from `DejitterCbr`.

1.14

- **Added**

- The field `identical` was added to `InputUdpSeamless`.

It is used to signal if the sources are expected to be real 2022-7.

23.2 Command Reference

23.2.1 GetInputs

Get a list of all configured inputs.

- message **GetInputs.Request**
- message **GetInputs.Response**

23.2.2 SetInputs

Update configuration for inputs. If an object appear in the request which is not already in the configuration database, it will be added. Otherwise it will be updated.

- message **SetInputs.Request**
- message **SetInputs.Response**

23.2.3 DeleteInputs

Delete inputs.

- message **DeleteInputs.Request**
- message **DeleteInputs.Response**

23.2.4 CheckUsageInputs

Check if a set of inputs are in use by any outputs

- message **CheckUsageInputs.Request**
- message **CheckUsageInputs.Response**

23.3 Type Reference

23.3.1 AdditionalInputFeatures

Additional Input Features.

struct

descrambling	optional bissDescrambling.DescramblerSettings Settings for descrambling of TS data. Only supports BISS version 2 and Fixed key with AES scrambling. This feature is only available on BISS products.
perFlowTriggers	optional triggers.TriggerConfig

23.3.2 CheckUsageInputs.Request

struct

ids	list of UUID
-----	----------------------------

23.3.3 CheckUsageInputs.Response

struct

data	map from UUID to bool
------	--

23.3.4 CheckUsageInputsRequest

struct

ids	list of UUID
-----	----------------------------

23.3.5 CheckUsageInputsResponse

struct

data	map from UUID to bool
------	--

23.3.6 Dejitter

Settings for dejitter.

struct

settings	DejitterSettings Contains a variant describing which dejitter version to use and its settings.
----------	--

bufferSize	int Buffer size given in milliseconds.
------------	--

23.3.7 DejitterCbr

Settings for constant bit rate dejittering.

struct

enableStatus	bool Undocumented in version 1.1. Always set to false.
useRtpTimestamps	bool Use RTP Timestamps for better dejittering.

23.3.8 DejitterPcr

Settings for PCR dejittering.

struct

preferredPcrPid	optional int Undocumented in version 1.1. Leave empty.
-----------------	--

23.3.9 DejitterRtp

Settings for RTP dejittering.

struct

enableStatus	bool Undocumented in version 1.1. Always set to false.
profile	RtpDejitterProfile Selects which RTP dejitter profile to use for dejittering

23.3.10 DejitterSettings

Dejitter settings variant. Allows selection between RTP dejitter, CBR dejitter, and PCR dejitter mode.

variant

rtp	DejitterRtp
cbr	DejitterCbr
pcr	DejitterPcr

23.3.11 DeleteInputs.Request

struct

ids	list of UUID
-----	---------------------

23.3.12 DeleteInputs.Response

empty **struct**

23.3.13 GetInputs.Request

empty **struct**

23.3.14 GetInputs.Response

struct

data	map from UUID to Input
info	firewallTypes.PaginationInfo

23.3.15 Input

An Input configuration opens the firewall for a given data stream. Configuration is split into two parts: transport and analysis. Transport configuration consists of IP protocol, or ASI settings, while analysis configuration consists of settings describing how the data should be interpreted by the system.

struct

label	string Human readable label for the input.
enabled	bool Whether the input is enabled or not.
transportSettings	InputTransportSettings Transport properties for this input.
analyzeMode	optional InputAnalyzeMode Analyze properties for this input.
additionalFeatures	optional AdditionalInputFeatures Some optional additional features, only supported on some products.

23.3.16 InputAnalyzeAtsc

Settings applying to an ATSC input.

struct

dejitter	optional Dejitter Dejitter settings for the input.
----------	--

23.3.17 InputAnalyzeDvb

Settings applying to a DVB input.

DVB Mode will analyze PID 0 (PAT) and PID 17 (SDT). The system will then analyze each PMT signaled in the PAT and announce it in the system for further processing, e.g. route to an output.

struct

dejitter

optional **Dejitter**

Dejitter settings for the input.

23.3.18 InputAnalyzeMode

Selecting the analyze mode and settings for the input source.

variant

mpegMode

InputAnalyzeMpeg

analyze the input as MPEG TS.

dvbMode

InputAnalyzeDvb

analyze the input as DVB TS.

rtpMode

InputRtp

do not analyze the input as TS.

atscMode

InputAnalyzeAtsc

do analyze the input as ATSC TS.

23.3.19 InputAnalyzeMpeg

Settings applying to an MPEG input.

struct

dejitter

optional **Dejitter**

Dejitter settings for the input.

23.3.20 InputRtp

Settings applying to an RTP input.

struct

dejitter

optional **Dejitter**

Dejitter settings for the input.

23.3.21 InputTransportSettings

InputTransportSettings determines which type of transport is used for this input.

variant

udp

InputUdpTransportSettings

If set, assumes this input is a UDP input and configures it using the provided settings.

asi

asiModule.InputAsiTransportSettings

If set, assumes this input is an ASI input and configures it using the provided settings.

srt

srt.SrtInputSettings

If set, assumes this input is an SRT input and configures it using the provided settings.

zixi.ZixiInputSettings

zixi	If set, assumes this input is an Zixi input and configures it using the provided settings.
------	--

23.3.22 InputUdp

UDP Input. May either be a single source or multiple in seamless mode (SMTPE-2022-7).

variant

seamless	InputUdpSeamless
single	InputUdpSettings

23.3.23 InputUdpSeamless

Set the input in seamless transport mode (SMTPE-2022-7). The two sources, a and b, must be setup at different interfaces.

struct

a	InputUdpSettings UDP settings for the A pipe.
b	InputUdpSettings UDP settings for the B pipe.
preferred	InputUdpSeamless.preferred Preferred pipe.
identical	bool Means the expected sources are 2022-7 compliant, i.e identical; If they are not a proper alarm will be raised
optimizeLatency	bool Ensure the 2022-7 buffer is optimized for latency. Resets the input if optimization is required.

23.3.24 InputUdpSeamless.preferred

enum

A
B
FLOATING

23.3.25 InputUdpSettings

UDP input settings. If destination is a multicast address IGMP join will be performed. A destination must be unique for a given interface.

struct

interfaceId	UUID Interface Id. See ipinterface.IpInterface
destination	firewallTypes.SocketAddress Unicast or multicast address
	bool

fec	FEC settings
rtp	bool Expect RTP header present, used when InputAnalyzeMode is rtpMode.
sources	list of string List of IGMP source addresses. Maximum number of sources allowed is 20.

23.3.26 InputUdpTransportSettings

InputUdpTransportSettings Settings for all UDP transport inputs

struct

input	InputUdp UDP input. Variant either single or seamless.
inputBufferSize	int Input buffer size. Buffer for reconstructing missing packets on the input. Ranges 1-400ms (default 20ms) for seamless and 300-400ms(default 300ms) for FEC. 0ms is set if neither FEC or seamless is active.
trafficPolicing	IpTrafficPolicing Traffic Policing settings.
rtpExcessiveLossThreshold	int Configurable threshold for the excessive RTP loss alarm (default 100).
alarmWhenScrambled	bool If set as true then enable alarm when input is scrambled, otherwise alarm is disable.

23.3.27 IpTrafficPolicing

Type to hold the traffic policing information for IP Inputs.

struct

enabled	bool When true, the input stream will be taken down if the input bitrate exceeds maxBitrate.
maxBitrate	int Max allowed bitrate for traffic policing, in units of (kbps). Allowed range is 0 - 213 mbps.

23.3.28 RtpDejitterProfile

The supported profiles for RTP dejittering

enum

SMPTE_2022_6	
SMPTE_2022_2	

23.3.29 SetInputs.Request

struct

data

map from **UUID** to **Input**

23.3.30 SetInputs.Response

empty **struct**

24 mappingTemplates (1.2)

24.1 Overview

Changelog:

1.2

- Changed
- Import componentMapping changed from version 1.2 to 1.3

1.1

- Changed
- Import mappingTemplates changed from version 1.1 to 1.2

24.2 Command Reference

24.2.1 GetTemplates

Gets templates stored in the unit

- message **GetTemplates.Request**
- message **GetTemplates.Response**

24.2.2 StoreTemplates

Stores templates on the unit

- message **StoreTemplates.Request**
- message **StoreTemplates.Response**

24.2.3 DeleteTemplates

Deletes templates on the unit

- message **DeleteTemplates.Request**
- message **DeleteTemplates.Response**

24.2.4 GetTemplateStatus

Get template status

- message **GetTemplateStatus.Request**
- message **GetTemplateStatus.Response**

24.3 Type Reference

24.3.1 AppliedTemplateRules

Keeps indices of applied template rules

list of **int**

24.3.2 DeleteTemplates.Request

struct

ids [list of UUID](#)

24.3.3 DeleteTemplates.Response

empty [struct](#)

24.3.4 GetTemplateStatus.Request

empty [struct](#)

24.3.5 GetTemplateStatus.Response

TemplateStatusResponse

24.3.6 GetTemplates.Request

empty [struct](#)

24.3.7 GetTemplates.Response

struct

data [map](#) from [UUID](#) to [OutputTemplates](#)

24.3.8 OutputTemplates

OutputTemplates aggregates all Component Mapping and Service Name templates.

struct

rules [list of TemplateRule](#)
List of handled templates.

24.3.9 ServiceMapping

ServiceMapping contains changes to service name in output mappings

struct

serviceName [string](#)
Name of mapped output service

24.3.10 StoreTemplates.Request

struct

data [map](#) from [UUID](#) to [OutputTemplates](#)

24.3.11 StoreTemplates.Response

empty **struct**

24.3.12 TemplateRule

TemplateRule contains matchers and mapping rules for output

struct

matchedNetworkId	optional gatewayValidate.IntRange Original Network ID range selector. If empty any Network ID will be matched.
matchedTransportId	optional gatewayValidate.IntRange Transport ID range selector. If empty any Transport ID will be matched.
matchedServiceId	optional gatewayValidate.IntRange Service ID range selector. If empty any Service ID will be matched.
mappings	TemplateRuleUpdates Rules to apply. Can be any type of supported template mappings.

24.3.13 TemplateRuleUpdates

TemplateRuleUpdates contains variant of either component or service mapping

variant

component	componentMapping.ComponentMapping Variant for component mappings
service	ServiceMapping Variant for service name mappings

24.3.14 TemplateStatus

TemplateStatus keeps track of applied rules (by its index) for particular sources in output

map from **UUID** to **list** of **AppliedTemplateRules**

24.3.15 TemplateStatusResponse

TemplateStatuses represents rules that have been applied to all outputs

struct

data	TemplateStatus
------	-----------------------

25 output (1.36)

25.1 Overview

Changelog:

1.36

- **Changed**
- The imported `srt` version was bumped to 1.6

1.35

- **Changed**
- The imported `bissScrambling` version was bumped to 1.6

1.34

- **Changed**
- The imported `srt` version was bumped to 1.5

1.33

- **Changed**
- The imported `srt` version was bumped to 1.4

1.32

- **Changed**
- The imported `srt` version was bumped to 1.3

1.31

- **Changed**
- The imported outputMapping version was bumped to 1.4

1.30

- **Changed**
- The imported outputMapping version was bumped to 1.3
- The imported bissScrambling version was bumped to 1.5

1.29

- **Changed**
- The imported srt version was bumped to 1.2

1.28

- **Changed**
- The imported outputMapping version was bumped to 1.2

1.27

- **Changed**
- The imported asiModule version was bumped to 1.11

1.26

- ****Changed****
- The imported asiModule version was bumped to 1.10

1.25

- ****Changed****
- The imported asimodule version was bumped to 1.9
- OutputMapping bumped to 1.1
- The imported `srt` version was bumped to 1.1.
- The imported `zixi` version was bumped to 1.1.
- remember output source address and port

1.24

- ****Changed****
- The imported asimodule version was bumped to 1.8
- OutputMapping was moved to its own idcl file

1.23

- **Changed**
- The imported bissScrambling version was bumped to 1.4.
- The imported asimodule version was bumped to 1.7

1.22

- **Added**
- The field scramblingIndex was added to
 - * DvbMultiplexServiceSettings
 - * MpegMultiplexProgramSettings
 - * AtscMultiplexChannelSettings
- The optional field additionalFeatures was added to Output.
 The AdditionalOutputFeatures type includes the field scrambling, which is used to configure scrambling on the output.
- Bumped asiModule from 1.6 to 1.7
- **Changed**
- The imported asiModule version was bumped to 1.7.
- The validation of the fields guardInterval and bufferDelay in InputRedundancy changed, making the intervals dependent on if the stream is CBR or VBR (in practice; whitelist/blacklist).
- Minimum allowed value of Max_offset is 4ms.
- Minimum allowed value of Guard_interval is 2ms when CBR and 50ms when VBR.
- Sum of Max_offset and Guard_interval may not be greater than 150ms.

1.21

- **Changed**
- The validation of the field switchbackDelay in InputRedundancy changed.
 Previous minimum value was 0, but has been increased to 10. Any values below 10 are set to 10.

- The validation of the fields `guardInterval` and `bufferDelay` in `InputRedundancy` changed for whitelist mode.

The minimum value for the guard interval was increased to 50, the maximum buffer delay is set to 100 and the maximum allowed sum of guard interval + buffer delay is set to 150. Any values outside the legal range are adjusted to be within the legal limits.

1.20

• Removed

- The field `scrambledComponents` was removed from
 - * `DvbMultiplexServiceSettings`
 - * `MpegMultiplexProgramSettings`
 - * `AtscMultiplexChannelSettings`

1.19

• Added

- The field `latencyTuning` was added to `IdenticalSources`.
This allows specifying if main is leading or lagging.
- The field `guardInterval` was added to `IdenticalSources`.
This allows specifying a delay used to compensate for fluctuations in the time offset between the flows.

25.2 Command Reference

25.2.1 GetOutputs

- message `GetOutputs.Request`
- message `GetOutputs.Response`

25.2.2 SetOutputs

- message `SetOutputs.Request`
- message `SetOutputs.Response`

25.2.3 DeleteOutputs

- message `DeleteOutputs.Request`
- message `DeleteOutputs.Response`

25.3 Type Reference

25.3.1 AdditionalOutputFeatures

Additional Output Features.

struct

bissScrambling.ScramblerSettings

scrambling

Settings for scrambling of TS data. Only supports BISS version 2 and Fixed key with AES scrambling. This feature is only available on BISS products.

25.3.2 DeleteOutputs.Request

struct

ids **list of UUID**

25.3.3 DeleteOutputs.Response

empty **struct**

25.3.4 FECMode

enum

A

B

25.3.5 FECSettings

struct

mode **FECMode**

columns **int**

rows **int**

25.3.6 GetOutputs.Request

variant

filter **GetOutputsFilter**

25.3.7 GetOutputs.Response

struct

data **map from UUID to Output**

A map of outputs returned by the request.

info **firewallTypes.PaginationInfo**

Meta information about the data - i.e the total number of items on the server.

25.3.8 GetOutputsFilter

struct

optional firewallTypes.PaginatedRequest

pagination

Settings for splitting the response data into pages

25.3.9 L2TPv3

L2TPv3 Configuration

struct

udp	bool
flowid	int
sessionid	bigint

25.3.10 Output

Output Configurations.

Output configuration is divided into two sections: transport and mapping. Transport settings consist of IP protocol or ASI settings, while mapping settings consist of settings related to stream content.

struct

label	string Name of the output
enabled	bool If the output is enabled or not
transportSettings	OutputTransportSettings Transport settings.
outputSettings	outputMapping.OutputMapping Mapping settings.
additionalFeatures	optional AdditionalOutputFeatures Some optional additional features, only supported on some products.

25.3.11 OutputTransportSettings

Variant for selecting the output transport settings

variant

udp	OutputUdpTransportSettings Used to configure UDP outputs
asi	asiModule.OutputAsiTransportSettings Used to configure ASI outputs
srt	srt.SrtOutputSettings Used to configure SRT outputs
zixi	zixi.ZixiOutputSettings Used to configure Zixi outputs

25.3.12 OutputUdp

UDP output variant - selects either cloned output or single output mode

variant

cloned	OutputUdpCloned Configuration for port A and port B
single	OutputUdpSettings Configuration for the only port used by this output

25.3.13 OutputUdpCloned

Cloned Configuration for port A and port B. Both interfaces must be configured to run at the same link speed.

struct

a	OutputUdpSettings
b	OutputUdpSettings

25.3.14 OutputUdpParameters

Parameters for UDP outputs

struct

dscp	int Type of service
ttl	int Time To Live
rtp	bool RTP
ts	optional OutputUdpTsParameters TS specific parameters
l2tpv3	optional L2TPv3 L2TPv3 settings
vlanPriority	optional VlanPriority

25.3.15 OutputUdpSettings

Output settings for UDP outputs.

struct

interfaceId	UUID
destination	firewallTypes.SocketAddress
sourceAddressEnabled	bool
sourceAddress	string
sourcePortEnabled	bool
sourcePort	int
fecEnabled	bool
fec	optional FECSettings

25.3.16 OutputUdpTransportSettings

Transport settings for an UDP output

struct

settings	OutputUdpParameters
output	OutputUdp UDP output settings – cloned or single port

25.3.17 OutputUdpTsParameters

UDP settings specific to TS

struct

mpegPacketsPerFrame	int How many MPEG packets are sent per transport frame
---------------------	--

25.3.18 SetOutputs.Request**struct**

data	map from UUID to Output
------	--

25.3.19 SetOutputs.Response

empty **struct**

25.3.20 VlanPriority

VlanPriority – Set priority related 802.1Q header flags

Value: Traffic types 0 Best effort 1 Background (lowest priority) 2 Excellent effort 3 Critical applications 4 Video, < 100 ms latency and jitter 5 Voice, < 10 ms latency and jitter 6 Internetwork control

struct

pcp	int
dei	bool

26 outputMapping (1.4)

26.1 Overview

Changelog:

1.4

- **Changed**
 - componentMapping bumped from 1.2 to 1.3
 - genericDescriptor bumped from 1.1 to 1.2

1.3

- **Added**
 - Added autoEmm option to TsSettings

1.2

- **Added**
 - Added floating switch mode to InputRedundancy

1.1

- **Changed**
 - genericDescriptor bumped to v 1.1

1.0

- **Added**
 - Added mapping part from output.1.24.idcl
 - The field `externalPids` was added to `OutputMappingTsBlacklist`

26.2 Type Reference

26.2.1 AnnexA

QAM Annex A input parameters, per the ITU-T J.83 specification.

By setting the Annex A parameters, one specifies how a TS should be QAMed, which ultimately results in a specific bitrate as described in the ITU-T spec.

struct

modulation	Modulation Specifies the QAM modulation scheme, either QAM64 or QAM256.
symbolrate	Symbolrate Symbol rate to use for QAM.

26.2.2 AnnexB

Annex B input parameters, per the ITU-T J.83 specification.

In contrast to Annex A, Annex B only has two choices for how to QAM, and therefore only has two possible output bitrates.

struct

modulation

Modulation

Specifies QAM modulation scheme, either QAM64 or QAM256. QAM64 results in a output bitrate of 26.970377 Mbps. QAM256 results in a output bitrate of 38.810700 Mbps.

26.2.3 AnnexC

QAM Annex C input parameters, per the ITU-T J.83 specification.

By setting the Annex C parameters, one specifies how a TS should be QAMed, which ultimately results in a specific bitrate as described in the ITU-T spec. Note that the resulting bitrate is the same as for Annex A.

struct

modulation

Modulation

Specifies the QAM modulation scheme, either QAM64 or QAM256.

symbolrate

Symbolrate

Symbol rate to use for QAM.

26.2.4 AtscChannelNumber

Major and minor channel overrides

struct

majorChannelNumber

int

Override major channel number.

minorChannelNumber

int

Override minor channel number.

26.2.5 AtscChannelSource

Source selection for a specific input service

struct

source

UUID

UUID for a input service.

inputRedundancy

InputRedundancy

priority

int

Value to identify the impact of losing specific source for redundancy logic

settings

AtscMultiplexChannelSettings

PID, PMT, CVCT and TVCT filters and overrides.

26.2.6 AtscMultiplexChannelSettings

PID, PMT, and SDT filters and overrides.

struct

programNumber	optional int
channelNumber	optional AtscChannelNumber Override channel number.
shortName	optional string Override short name.
componentLayout	componentMapping.ComponentLayoutMap Remap, copy, and filter PIDs and components.
pmtPid	optional int Override PMT PID.
componentGeneration	list of componentMapping.ComponentGeneration
descriptors	list of genericDescriptor.GenericPmtDescriptorEntry Add, change or remove descriptors on this service.
scramblingIndex	optional int Scrambling index. May be given if scrambling is enabled for the TS, and this services should have different settings than the default settings.

26.2.7 AtscMultiplexSource

ATSC multiplex source type, which must be either be a channel, an EMM or an external PID. An empty AtscMultiplexSource is considered an invalid config.

variant

channel	AtscChannelSource An ATSC channel (service)
emm	EmmSource An EMM source
externalPid	ExternalPidMapping An External PID

26.2.8 AtscSource

variant

multiplex	list of AtscMultiplexSource
-----------	------------------------------------

26.2.9 AtscTableSettings

ATSC PSI Table Settings

Settings to choose payout of PSI tables

struct

cvct	bool Set payout of CVCT enabled/disabled
tvct	bool Set payout of TVCT enabled/disabled

26.2.10 Bitrate

Bitrate describes the bitrate scheme to be used. Only one of the parameters can be specified.

variant

manual	int An integer that sets the desired bitrate in kbps (max 3000Mbps).
annexA	AnnexA A special struct for ITU-T J.83 Annex A, which contains modulation type and symbol rate.
annexB	AnnexB A special struct for ITU-T J.83 Annex B, which contains modulation type.
annexC	AnnexC A special struct for ITU-T J.83 Annex C, which contains modulation and symbol rate.
vsb8	Vsb8 Used to indicate that the specific VSB-8 bitrate is to be used.

26.2.11 DvbMultiplexServiceSettings

PID, PMT, and SDT filters and overrides.

struct

serviceId	optional int Override service ID.
serviceName	optional string Override service name.
serviceType	optional int Override service type.
serviceProvider	optional string Override service provider.
componentLayout	componentMapping.ComponentLayoutMap Remap, copy, and filter PIDs and components.
pmtPid	optional int Override PMT PID.
componentGeneration	list of componentMapping.ComponentGeneration
descriptors	list of genericDescriptor.GenericPmtDescriptorEntry Add, change or remove descriptors on this service.
scramblingIndex	optional int Scrambling index. May be given if scrambling is enabled for the TS, and this service should have different settings than the default settings.

26.2.12 DvbMultiplexSource

DVB multiplex source type, which must be either be a service, an EMM or an external PID. An empty DvbMultiplexSource is considered an invalid config.

variant

service	DvbServiceSource An DVB service
emm	EmmSource An EMM source
externalPid	ExternalPidMapping An External PID

26.2.13 DvbServiceSource

Source selection for a specific input service

struct

source	UUID UUID for a input service. The service ID can be queried using services:l.0/GetInputServices or calculated: UUID5(Input UUID, "dvbsource service SERVICEID").
inputRedundancy	InputRedundancy is Input Redundancy configuration for this output
priority	int Value to identify the impact of losing specific source for redundancy logic
settings	DvbMultiplexServiceSettings PID, PMT, and SDT filters and overrides.

26.2.14 DvbSource

variant

multiplex	list of DvbMultiplexSource
-----------	--

26.2.15 EmmSource

EMM source

struct

source	UUID UUID of an EMM
inputRedundancy	InputRedundancy is Input Redundancy configuration for this output
priority	int Value to identify the impact of losing specific source for redundancy logic

26.2.16 ExternalPidMapping

Map specific pid from a source with the specified uuid to specific destination pid on output

struct

UUID

source	UUID to identify which input source to map pid from
priority	int Value to identify the impact of losing specific source for redundancy logic
srcPid	int PID to map from the specified source
dstPid	int PID to map the source PID to on the output

26.2.17 HotStandby

struct

identicalSources	optional IdenticalSources When set MAIN and first BACKUP sources will be switched internally in FPGA module. MAIN and first BACKUP sources should have identical contents.
------------------	--

26.2.18 IdenticalSources

struct

bufferDelay	int Delay in milliseconds for offset handling between streams.
guardInterval	int Delay in milliseconds used to compensate for fluctuations in the time offset between the flows
switchOnTei	bool When enabled FPGA module additionally uses TEI status to assess alternative health. Available only for HotStandbyWith-IdenticalSources switch mode. Only one alternative allowed.
latencyTuning	LatencyTuning Selected latency tuning strategy

26.2.19 InputRedundancy

Data structure related to Input Redundancy feature

struct

mode	InputRedundancySwitchMode Selected switch mode
backups	list of UUID List of alternative sources
preferredSource	optional PreferredSource When set allows to override default source preferences
switchbackDelay	int Delay in seconds to wait before switching back to a more preferred alternative or MAIN source.

optional HotStandby

hotStandby

When set all alternative's bitrates will be monitored. Flows are preemptively mapped to the output card providing improved switch times.

26.2.20 InputRedundancySwitchMode

Enum related to SwitchMode between alternative sources

enum

Disabled	InputRedundancy is inactive; only MAIN source is used.
Reverting	Always switch to the highest priority, healthy alternative. Usually after it has been healthy for a set SWITCHBACK DELAY. Main has the highest priority, then backup 1, backup 2, etc.
Floating	As opposed to reverting, it doesn't go back to a higher priority alternative even if it becomes healthy again after a SWITCHBACK DELAY.

26.2.21 LatencyTuning

Enum describing desired latency tuning strategy

enum

Auto	The first flow activated will be assumed to be leading
MainLeading	The Main source is assumed to be leading
MainLagging	The Main source is assumed to be lagging

26.2.22 Modulation

QAM Modulation Types.

The J.83 specification utilizes two QAM variants, QAM64 and QAM256. It is denoted as "64-QAM" and "256-QAM" in the ITU-T J.83 spec. The number denotes the amount of points in the signal constellation used when performing QAM.

enum

QAM64
QAM256

26.2.23 MpegMultiplexProgramSettings

PID and PMT filters and overrides.

struct

programNumber	optional int Override program number (Equivalent to serviceId in DVB).
componentLayout	componentMapping.ComponentLayoutMap Remap, copy and filter PIDs and components.
pmtPid	optional int Override the PMT PID.
componentGeneration	list of componentMapping.ComponentGeneration

descriptors	list of genericDescriptor.GenericPmtDescriptorEntry Add, change or remove descriptors on this service.
scramblingIndex	optional int Scrambling index. May be given if scrambling is enabled for the TS, and this services should have different settings than the default settings.

26.2.24 MpegMultiplexSource

MPEG multiplex source type, which must be either be a program, an EMM or an external PID. An empty MpegMultiplexSource is considered an invalid config.

variant

program	MpegProgramSource An MPEG program (service)
emm	EmmSource An EMM source
externalPid	ExternalPidMapping An External PID

26.2.25 MpegProgramSource

MPEG program source and settings bundle.

struct

source	UUID UUID of an input program (MPEG equivalent of DVB service)
inputRedundancy	InputRedundancy is Input Redundancy configuration for this output
priority	int Value to identify the impact of losing specific source for redundancy logic
settings	MpegMultiplexProgramSettings PID and PMT filters and overrides for this program.

26.2.26 MpegSource

An MPEG mode source. Can be SPTS or MPTS, where SPTS is considered a single program multiplex struct.

variant

multiplex	list of MpegMultiplexSource A list of MPEG sources this output consists of.
-----------	--

26.2.27 OutputMapping

Output settings variant, contains settings for the supported output types. It must be either tsWhitelistMode, tsBlacklistMode or rawMode. An empty OutputMapping is considered an invalid config.

variant

tsWhitelistMode	OutputMappingTsWhitelist Build TS by adding content
tsBlacklistMode	OutputMappingTsBlacklist Map entire TS, and possibly exclude part of the stream
rawMode	OutputMappingRaw Transparently map entire flow

26.2.28 OutputMappingAtsc

ATSC MPTS/SPTS Settings.

The system will generate PAT, PMT, MGT, and possibly CVCT and TVCT

struct

settings	TsSettings Generic stream settings.
source	AtscSource Content to multiplex.
tables	AtscTableSettings PSI table settings

26.2.29 OutputMappingDvb

DVB MPTS/SPTS Settings.

The system will generate PAT, PMT, and SDT

struct

orgNetworkId	optional int Orginal Network Id to set on stream.
settings	TsSettings Generic stream settings.
source	DvbSource Content to multiplex.

26.2.30 OutputMappingMpeg

struct

settings	TsSettings
source	MpegSource

26.2.31 OutputMappingRaw

Mapping of stream as is

struct

UUID

source	Main source of stream
inputRedundancy	InputRedundancy Input Redundancy configuration for this output

26.2.32 OutputMappingTsBlacklist

Transport stream blacklist mapping

If CBR is specified, the original NULL PIDs are discarded.

struct

source	UUID Main TS source
inputRedundancy	InputRedundancy Input Redundancy configuration for this output
bitrate	Bitrate Variant Bitrate – if specified, the transport is CBR, AnnexA or AnnexB with relevant attributes. Otherwise, it is VBR.
stopPids	list of int PIDs to be stopped on Payout
remapPids	map from int to int PIDs to be remapped on Payout
externalPids	list of ExternalPidMapping List of PID mappings from external sources

26.2.33 OutputMappingTsWhitelist

Transport stream whitelist mapping which must be either DVB, MPEG or ATSC. An empty OutputMappingTsWhitelist is considered an invalid config.

variant

mpegMode	OutputMappingMpeg MPEG MPTS/SPTS settings
dvbMode	OutputMappingDvb DVB MPTS/SPTS settings
atscMode	OutputMappingAtsc ATSC MPTS/SPTS settings

26.2.34 PreferredSource

Preferred source configuration

struct

index	int Selects a source that should be marked as preferred. Index = 0 is MAIN source, index = 1 means first backup, etc. Preferred source will be used as the default source when it is healthy. If it fails the system will switch using regular source priorities (for example: first MAIN, then first backup, then second backup and so on)
-------	---

locked	bool When enabled disables switching to another source when preferred source is faulty
--------	--

26.2.35 Symbolrate

Symbol rate.

A symbol rate is defined in DOCSIS 2.0 as $(m/n) * 10.24\text{MHz}$, where M and N are 16-bit unsigned, nonzero integers, and 10.24MHz is the CMTS clock frequency.

struct

m	int The numerator of the desired fraction of CMTS clock frequency.
n	int The denominator of the desired fraction of CMTS clock frequency.

26.2.36 TsSettings

Settings shared between all TS configurations

struct

tsId	int Transport stream ID for the output TS
bitrate	Bitrate Variant Bitrate - if specified, the transport is CBR, AnnexA or AnnexB with relevant attributes. Otherwise, it is VBR.
autoEmm	bool Automatically include CAT and EMM from the input source. Automatic CAT/EMM is supported for single service outputs (SPTS) only. Also, only the CAS ID referenced in the ECM for the service will be mapped. The CAT will be regenerated to signal the required CAS ID only.

26.2.37 Vsb8

8-VSB struct. It is used by the Bitrate type to specify that the 8-VSB modulation scheme is desired. Per the specification, 8-VSB has no configurable inputs, and returns a MPEG II stream of 19.39...Mbits/s ready for 8-VSB modulation.

empty **struct**

27 pagination (1.0)

27.1 Type Reference

27.1.1 DataPageInfo

struct

startIndex	int
hasReachedEnd	bool

27.1.2 DataSetInfo

struct

token	optional UUID Used to inform about changes in dataset. If backend can detect changes in dataset then this should have a value. If this value changes it means that dataset has changed.
size	optional int Total size of full dataset. Should have a value if backend can know the size of full dataset.

27.1.3 DataSetSlice

struct

startIndex	int
elementCount	int

27.1.4 PaginatedDataInfo

struct

datasetInfo	DataSetInfo
pageInfo	DataPageInfo

27.1.5 PaginatedQueryResult

variant

ok	PaginatedDataInfo Requested data can be addressed and you get it now.
processing	PaginatedQueryResult.processing Requested data can be addressed but you can't get it now, system started to retrieve the data so you will be able to get it soon.
noContent	PaginatedQueryResult.noContent Requested data can be addressed but you can't get it now, system is in a state in which it's currently impossible to retrieve data.

27.1.6 PaginatedQueryResult.noContent

empty **struct**

27.1.7 PaginatedQueryResult.processing

empty **struct**

28 services (1.3)

28.1 Command Reference

28.1.1 GetInputServices

Returns service information for the queried list of service node IDs. By default returns only services which are currently present on inputs. Can be provided a query object which can change default behavior.

- message `GetInputServices.Request`
- message `GetInputServices.Response`

28.1.2 GetInputServicesPage

- message `GetInputServicesPage.Request`
- message `GetInputServicesPage.Response`

28.1.3 GetOutputServices

Returns service information for output services

- message `GetOutputServices.Request`
- message `GetOutputServices.Response`

28.1.4 GetServiceDependencies

Returns service dependency information

- message `GetServiceDependencies.Request`
- message `GetServiceDependencies.Response`

28.2 Type Reference

28.2.1 ChildSourceInfo

struct

name	string
nodeType	string
body	string
owner	string
isPresent	bool

28.2.2 FlowSinkInfo

struct

name	string
label	string
nodeType	string
body	string

sources	map from UUID to FlowSourceServiceInfo
owner	string

28.2.3 FlowSourceInfo

struct

name	string
label	string
nodeType	string
body	string
owner	string
isPresent	bool
sources	map from UUID to ChildSourceInfo

28.2.4 FlowSourcePage

struct

result	pagination.PaginatedQueryResult
data	map from UUID to FlowSourceInfo

28.2.5 FlowSourceServiceInfo

ChildSourceInfo

28.2.6 GetInputServices.Request

GetInputServicesRequest

28.2.7 GetInputServices.Response

struct

data	map from UUID to FlowSourceInfo
------	---------------------------------

28.2.8 GetInputServicesPage.Request

GetInputServicesPageRequest

28.2.9 GetInputServicesPage.Response

FlowSourcePage

28.2.10 GetInputServicesPageRequest

struct

slice	pagination.DataSetSlice
query	optional InputServiceQuery

28.2.11 GetInputServicesRequest

struct

query	optional InputServiceQuery
-------	----------------------------

28.2.12 GetOutputServices.Request

empty struct

28.2.13 GetOutputServices.Response

struct

data	map from UUID to FlowSinkInfo
------	-------------------------------

28.2.14 GetServiceDependencies.Request

empty struct

28.2.15 GetServiceDependencies.Response

struct

data	list of ServiceDependencyInfo
------	-------------------------------

28.2.16 InputServicePresenceQuery

enum

ALL
ONLY_PRESENT
ONLY_MISSING

28.2.17 InputServiceQuery

struct

servicePresence	InputServicePresenceQuery
-----------------	---------------------------

28.2.18 ServiceDependencyInfo

struct

source	UUID
dependency	UUID

29 srt (1.6)

29.1 Overview

Changelog:

1.6

- **Added**
 - Added 'maxNumberOfConnections' to SrtOutputSettings
 - Added 'rejectedPeers' and 'acceptedPeers' to SrtListener
 - Added 'streamIdFilter' to SrtListener
- **Changed**
 - Changed max overhead bandwidth from 100
 - Changed min latency from 60ms to 1ms
 - Moved streamId from SrtOutputSettings and SrtInputSettings to SrtCaller

1.5

- **Added**
 - Added 'trafficPolicing' to both SrtOutputSettings and SrtInputSettings

1.4

- **Added**
 - Added 'highBitrate' to SrtOutputSettings and SrtInputSettings

1.3

- **Added**
 - Added 'streamId' to SrtOutputSettings and SrtInputSettings
- **Changed**
 - Option 'AES_192' has been removed from SrtAlgorithm
 - Port parameters in SrtCaller have changed names. It now has the same meaning independent of input / output.
 - Port parameter in SrtListener also have changed name for consistency with caller.

1.2

- **Added**
 - Added field 'rtp' to type SrtOutputSettings

1.1

- **Added**
 - Added the field 'transport' to types SrtOutputSettings and SrtInputSettings
 - Added the types SrtInputTransport, SrtOutputTransport, SrtSeamlessTransport, SrtClonedTransport and SrtTransportSettings
- **Moved**
 - Moved the fields 'interfaceld' and 'mode' from SrtOutputSettings and SrtInputSettings to

SrtTransportSettings

29.2 Type Reference

29.2.1 SrtAlgorithm

SRT Encryption Algorithm

enum

AES_128	AES with 128-bit key
AES_256	AES with 256-bit key

29.2.2 SrtCaller

SRT Caller Mode

struct

address	string IP address of remote / peer
localPort	int Local UDP port
remotePort	int Remote / peer UDP port
streamId	optional string SRT Stream ID

29.2.3 SrtClonedTransport

SRT Cloned Transport

struct

pathA	SrtTransportSettings Cloned Path A
pathB	SrtTransportSettings Cloned Path B

29.2.4 SrtEncryption

SRT Encryption

struct

algorithm	SrtAlgorithm Encryption algorithm
key	string Encryption key as a hex string

29.2.5 SrtInputSettings

SRT Input Settings

struct

transport	SrtInputTransport SRT input transport settings.
decryption	optional SrtEncryption Decryption settings. Not applicable for listener mode.
receiveLatency	int Receive latency in milliseconds.
highBitrate	bool Enabling high bitrate (HBR) mode will allow the stream to use higher maximum bitrate.
trafficPolicing	optional int Traffic policing of incoming data at given data rate (in kbps). If the input data rate is above the configured value, then the input is muted.

29.2.6 SrtInputTransport

SRT Input Transport

Seamless or single input

variant

seamless	SrtSeamlessTransport Seamless input
single	SrtTransportSettings Single input

29.2.7 SrtListener

SRT Listener Mode

struct

localPort	int Local UDP port
rejectedPeers	optional list of string List of rejected peers, using CIDR notation
acceptedPeers	optional list of string List of accepted peers, using CIDR notation
streamIdFilter	optional string Incoming connections will be rejected if their stream ID does not matches this value

29.2.8 SrtMode

SRT Mode

variant

caller	SrtCaller Caller mode
listener	SrtListener Listener mode
rendezvous	SrtRendezvous Rendezvous mode

29.2.9 SrtOutputSettings

SRT Output Settings

struct

transport	SrtOutputTransport SRT output transport settings
tsPacketsPerFrame	int Number of TS packets per IP frame [1-7]
encryption	optional SrtEncryption Encryption settings. Not applicable for listener mode.
receiveLatency	int Receive latency in milliseconds
overheadBandwidth	int Overhead bandwidth in per cent
rtp	bool Send TS with RTP headers (needed if received as 2022-7)
highBitrate	bool Enabling high bitrate (HBR) mode will allow the stream to use higher maximum bitrate.
trafficPolicing	optional int Traffic policing of outgoing data at given data rate (in kbps). If the source data rate is above the configured value, then the output is muted.
maxNumberOfConnections	optional int Maximum number of connections that can be established, only applicable for listener mode.

29.2.10 SrtOutputTransport

SRT Output Transport

Cloned or single output

variant

cloned	SrtClonedTransport Cloned output
single	SrtTransportSettings Single output

29.2.11 SrtRendezvous

SRT Rendezvous Mode

struct

address	string IP address of remote / peer
port	int UDP port (same for both local and remote)

29.2.12 SrtSeamlessTransport

SRT Seamless Transport

struct

pathA	SrtTransportSettings Seamless Path A
pathB	SrtTransportSettings Seamless Path B
preferred	SrtSeamlessTransport.preferred Preferred pipe
identical	bool Means the expected sources are 2022-7 compliant
seamlessBufferSize	int Input buffer size. Buffer for reconstructing missing packets on the input. Ranges 1-400ms

29.2.13 SrtSeamlessTransport.preferred**enum**

A	
B	
FLOATING	

29.2.14 SrtTransportSettings

SRT Transport Settings

struct

interfaceId	UUID UUID of the network interface
mode	SrtMode SRT mode

30 srtCardSettings (1.0)

30.1 Command Reference

30.1.1 GetCardSettings

- message **GetCardSettings.Request**
- message **GetCardSettings.Response**

30.1.2 SetCardSettings

- message **SetCardSettings.Request**
- message **SetCardSettings.Response**

30.2 Type Reference

30.2.1 CardSettings

Global settings for the SRT module

struct

maxSrtLatency

MaxSrtLatency

Global maximum SRT latency. Higher maximum SRT latency will reduce the maximum total bitrate, and also the maximum bitrate per stream.

30.2.2 GetCardSettings.Request

empty **struct**

30.2.3 GetCardSettings.Response

struct

config

CardSettings

30.2.4 MaxSrtLatency

Maximum SRT latency

enum

SECONDS_1	Maximum 1 second SRT latency
SECONDS_2	Maximum 2 second SRT latency
SECONDS_4	Maximum 4 second SRT latency
SECONDS_6	Maximum 6 second SRT latency
SECONDS_8	Maximum 8 second SRT latency

30.2.5 SetCardSettings.Request

struct

config **CardSettings**

30.2.6 SetCardSettings.Response

empty **struct**

31 srtValidate (1.1)

31.1 Type Reference

31.1.1 FloatRange

Float Range

struct

min	float Minimum value of the range
max	float Maximum value of the range

31.1.2 IntRange

Integer Range

struct

min	int Minimum value of the range
max	int Maximum value of the range

31.1.3 Minimum

Minimum

struct

min	int Minimum value of the range
-----	--

31.1.4 Result

Validation result

struct

result	bool True if config is valid and false otherwise.
msg	optional string An error message, which may be given if the result is invalid.

32 status (1.24)

32.1 Overview

Changelog:

1.24

- **Added**
 - added `srtSocketId` to `'SrtConnectionStatus'`
 - Renamed previous `'SrtOutputStatus'` to `'SrtOutputConnectionStatusPair'` and created a new `SrtOutputStatus` with list of `'SrtOutputStatusPair'`, `'activeConnections'` and `'connectionCountTotal'`

1.23

- **Added**
 - The field `'rtpErrors'` was added to `'OutputStatus'`

1.22

- **Added**
 - Additional SRT connection status fields; `'peerEndpoint'`, `'peerStreamId'`, `'srtLatency'`, and `'uptime'`.
- **Changed**
 - `'SrtSocketState'` has reduced number of enum values, removing unused values. Values are also renamed.
 - Formats of `'SrtInputConnectionStatus'` and `'SrtOutputConnectionStatus'` are slightly changed, moving connection related values into common type `'SrtConnectionStatus'`

1.21

- **Added**
 - The field `'bufferDelay'` was added to the `'SeamlessPortStatus'` type.

1.20

- **Removed**
 - Rpc `'GetOutputPidStatus'` has been moved to `tsstatus`, and thus has been removed from this file.

1.19

- **Added**
 - The type `'IpSourceAddressStatus'` where added.
- **Changed**
 - `'ipSourceAddressStatus'` optional was added on type `'IpInputStatus'`.

1.18

- **Added**
 - The field `'totalFlowBitrateWithoutFec'` was added to `'IpInputStatus'`
- **Changed**
 - `'totalFlowBitrate'` was changed from float to bigint in `'IpInputStatus'`, and the unit changed

from Kbps to bps.

- ‘effectiveFlowBitrate’ was changed from float to bigint in ‘IpInputStatus’, and the unit changed from Kbps to bps.
- ‘bitrate’ was changed from float to bigint in ‘SeamlessPortStatus’, and the unit changed from Kbps to bps.

1.17

• Changed

- ‘sequenceErrors’ was changed from int to bigint in ‘SeamlessPortStatus’
- ‘rtPErrors’ was changed from int to bigint in ‘IpInputStatus’

1.16

• Added

- Add GetOutputPidStatus RPC
- The types ‘ZixiInputPathStatus’ and ‘SrtInputPathStatus’ were added.
- The fields ‘inbandLinkStatus’ and ‘zixiStatus’/‘srtStatus’ were moved into the field ‘path1’ on types ‘ZixiInputStatus’ and ‘SrtInputStatus’.
- The fields ‘rtPErrors’, ‘syncByteError’, ‘teiBitError’, ‘dejitterStatus’ and ‘seamlessStatus’ were added to types ‘ZixiInputStatus’ and ‘SrtInputStatus’.
- The field ‘srtStatus’ was renamed to ‘path1’ on type ‘SrtOutputStatus’.
- The field ‘zixiStatus’ was renamed to ‘path1’ on type ‘ZixiOutputStatus’.
- The field ‘path2’ was added to types ‘ZixiOutputStatus’, ‘SrtOutputStatus’, ‘ZixiInputStatus’ and ‘SrtInputStatus’.

• Changed

- The imported version of tsstatus was bumped to 1.3

1.15

• Added

- The field ‘IpOutputBitrates’ was added to OutputStatus.

1.14

• Added

- The field ‘totalFlowBitrate’ and ‘totalFlowBitsReceived’ was added to IpInputStatus

• Changed

- The field InbandLinkStatus.statusCcErrors has been renamed to InbandLinkStatus.seqErrorCount

1.13

• Added

- The field alternativeStatuses was added to OutputSourceStatus.

• Changed

- The imported version of tsstatus was bumped to 1.1.
The fields tsId and originalNetworkId were added to TransportStreamStatus.

1.12

- Added

- The fields `mainBufferDelay`, `backupBufferDelay`, and `currentAlternativeIdx` were added to `OutputSourceStatus`.

32.2 Command Reference

32.2.1 GetIpInputStatus

- message `GetIpInputStatus.Request`
- message `GetIpInputStatus.Response`

32.2.2 ClearAllCounters

- message `ClearAllCounters.Request`
- message `ClearAllCounters.Response`

32.2.3 GetPidStatus

Get Pid status for a set of inputs.

- message `GetPidStatus.Request`
- message `GetPidStatus.Response`

32.2.4 GetZixiInputStatus

Get zixi status for a set of inputs.

- message `GetZixiInputStatus.Request`
- message `GetZixiInputStatus.Response`

32.2.5 GetZixiOutputStatus

Get zixi status for a set of outputs.

- message `GetZixiOutputStatus.Request`
- message `GetZixiOutputStatus.Response`

32.2.6 GetSrtInputStatus

- message `GetSrtInputStatus.Request`
- message `GetSrtInputStatus.Response`

32.2.7 GetSrtOutputStatus

- message `GetSrtOutputStatus.Request`
- message `GetSrtOutputStatus.Response`

32.2.8 GetOutputStatus

- message `GetOutputStatus.Request`
- message `GetOutputStatus.Response`

32.3 Type Reference

32.3.1 AlternativeStatus

Input Redundancy alternative status

struct

isBroken	bool
	indicates status of an alternative

32.3.2 CbrRegulationStage

Enum to hold possible stages of CBR dejittering algorithm

enum

INIT_FILTER_STAGE
FIRST_FILTER_STAGE
SECOND_FILTER_STAGE
REGULATION_STAGE

32.3.3 ClearAllCounters.Request

empty **struct**

32.3.4 ClearAllCounters.Response

empty **struct**

32.3.5 DejitterStatus

Type to hold status information for input dejittering

struct

mode	DejitteringMode
stage	optional CbrRegulationStage
maxJitter	float
minDelay	float
resetCounter	int

32.3.6 DejitteringMode

Enum to hold possible dejittering modes, status for PCR dejittering is not supported.

enum

CBR
RTP
PCR
OFF

32.3.7 FECStatus

struct

received	bool
enabled	bool
fecType	FECType
columns	int
rows	int
invalidParams	bool
decTimeout	bool
delayToLow	bool
unrecvBurstLoss	bool
columnRate	int
rowRate	int
unrecoverable	int
recovered	int
sequenceErrors	int
recommendedBufferSize	int
totalBitrateWithFEC	float

32.3.8 FECType

Fec type A Column only, Fect type B Row and Column

enum

A	
B	

32.3.9 GetIpInputStatus.Request

IpStatusReq

32.3.10 GetIpInputStatus.Response

IpStatusRes

32.3.11 GetOutputStatus.Request

empty struct

32.3.12 GetOutputStatus.Response

GetOutputStatusResponse

32.3.13 GetOutputStatusResponse

struct

data	map from UUID to OutputStatus
------	-------------------------------

32.3.14 GetPidStatus.Request

PidStatusRequest

32.3.15 GetPidStatus.Response

PidStatusResponse

32.3.16 GetSrtInputStatus.Request

SrtStatusReq

32.3.17 GetSrtInputStatus.Response

SrtStatusInputRes

32.3.18 GetSrtOutputStatus.Request

SrtStatusReq

32.3.19 GetSrtOutputStatus.Response

SrtStatusOutputRes

32.3.20 GetZixiInputStatus.Request

ZixiInputStatusReq

32.3.21 GetZixiInputStatus.Response

ZixiInputStatusRes

32.3.22 GetZixiOutputStatus.Request

ZixiOutputStatusReq

32.3.23 GetZixiOutputStatus.Response

ZixiOutputStatusRes

32.3.24 InbandLinkStatus

In-band Link Status

struct

sourceHostname	string Hostname of source module
sourceLabel	string User label of source flow

delay	optional float End-to-end packet delay (ms), 0.032ms resolution up to 2.115 s
jitter	float Packet jitter (us), 12.8 us resolution up to 0.84 s
seqErrorCount	bigint Number of sequential errors for the status packets
timeSourceSender	TimeSource Time source used on the sender
timeSourceReceiver	TimeSource Time source used on the receiver

32.3.25 IpInbandLinkStatus

In-band Link Status for IP Inputs

struct

path1	optional InbandLinkStatus Status for single input, or status for input path 1 if input is configured as seamless.
path2	optional InbandLinkStatus Status for input path 2 if input is configured as seamless.

32.3.26 IpInputBitrates

struct

totalFlowBitrate	bigint
totalFlowBitsReceived	bigint
effectiveFlowBitrate	bigint
effectiveFlowBitsReceived	bigint
totalFlowWithoutFecBitrate	bigint
totalFlowWithoutFecBitsReceive	bigint

32.3.27 IpInputStatus

struct

hasBitrate	bool
bitrates	optional IpInputBitrates
rtpErrors	optional bigint
ccErrors	optional int
syncByteError	optional int
teiBitError	optional int
services	list of ServiceSourceInfo
fecStatus	optional FECStatus
seamlessStatus	optional SeamlessStatus
dejitterStatus	optional DejitterStatus
inbandLinkStatus	IpInbandLinkStatus

ipSourceAddressStatus	IpSourceAddressStatus
-----------------------	------------------------------

32.3.28 IpOutputBitrates

struct

total	IpOutputTotalBitrates
effective	IpOutputEffectiveBitrates

32.3.29 IpOutputEffectiveBitrates

struct

effectiveFlowBitrate	float
effectiveFlowBitsSent	bigint

32.3.30 IpOutputTotalBitrates

struct

totalFlowBitrate	float
totalFlowBitsSent	bigint

32.3.31 IpSeamlessSourceStatus

Seamless ip source status

struct

a	optional IpSourceStatus
b	optional IpSourceStatus

32.3.32 IpSourceAddressStatus

Seamless or Single ip source status

variant

seamless	IpSeamlessSourceStatus
single	optional IpSourceStatus

32.3.33 IpSourceStatus

Input ip source address status

struct

multipleSources	bool
sourceIpAddress	string

32.3.34 IpStatusReq

variant

ids **list of UUID**

32.3.35 IpStatusRes

struct

data **map from UUID to IpInputStatus**

32.3.36 OutputSourceStatus

GetOutputStatus : Units are all in microseconds (us), but in the GUI it is converted to milliseconds (ms) with two decimals.

struct

effectiveIdenticalSourceOffset	optional int us of delay between Hot MAIN and BACKUP sources; it will be filled if OutputSource is configured with HotStandbyWithIdenticalSources
mainBufferDelay	optional int Current buffer delay in microseconds (us) for MAIN source. Undefined when MAIN is faulty
backupBufferDelay	optional int Current buffer delay in us for BACKUP source. Undefined when BACKUP is faulty
currentAlternativeIdx	int Index of currently selected alternative. 0 is MAIN, 1 is first backup and so on
alternativeStatuses	list of AlternativeStatus Status of each alternative defined for given source

32.3.37 OutputStatus

struct

bitrates	IpOutputBitrates
outputSources	list of OutputSourceStatus list of output's outputSources
rtpErrors	optional bigint The amount of RTP errors seen on this output

32.3.38 PidStatusRequest

struct

inputs **list of UUID**

32.3.39 PidStatusResponse

struct

data `map` from `UUID` to `tsstatus.TransportStreamStatus`

32.3.40 SeamlessPortStatus

struct

activeSource	<code>firewallTypes.SocketAddress</code>
bitrate	<code>bigint</code>
seamlessProtocol	<code>SeamlessProtocol</code>
role	<code>SeamlessRole</code>
sequenceErrors	<code>bigint</code>
bufferDelay	<code>optional int</code>

32.3.41 SeamlessProtocol

enum

RTP	
SequencedUDP	
UDP	
Unknown	

32.3.42 SeamlessRole

enum

Master	
Slave	

32.3.43 SeamlessStatus

struct

portA	<code>optional SeamlessPortStatus</code>
portB	<code>optional SeamlessPortStatus</code>
relativeDelay	<code>optional float</code>
synchronized	<code>bool</code>

32.3.44 ServiceSourceInfo

struct

serviceId	<code>int</code>
name	<code>string</code>
owner	<code>string</code>

32.3.45 SrtConnectionStatus

SRT Connection Status

Status for an active SRT connection.

struct

srtSocketId	int Local SRT socket ID
peerEndpoint	string UDP endpoint of peer, given as a string of the form 'ip:port'.
peerStreamId	optional string SRT Stream ID reported by peer. Status is only applicable for listeners, as only callers can report Stream ID.
srtLatency	int Actual SRT latency of the connection, as a result of handshake, given in milliseconds.
rtt	float Round trip time, given in milliseconds (ms).
uptime	int Connection uptime, given in seconds.

32.3.46 SrtInputConnectionStatus

SRT Input Connection Status

struct

socketStatus	SrtSocketState Current SRT socket status
connectionStatus	optional SrtConnectionStatus SRT connection status, if connected
totalPackets	bigint Total number of received packets
totalReceivedRetransmittedPackets	bigint Total number of received retransmitted packets
totalDroppedPackets	bigint Total number of packets dropped

32.3.47 SrtInputPathStatus

SRT Input Path Status Status for input if the input is single, or one path if input is seamless

struct

srtStatus	optional SrtInputConnectionStatus SRT status for path
inbandLinkStatus	optional InbandLinkStatus Inband Link Status for path

32.3.48 SrtInputStatus

SRT Input Status

struct

hasBitrate	bool True if input TS has bitrate
rtpErrors	optional int RTP Errors on input
syncByteError	optional int Sync Byte Errors on input
teiBitError	optional int TEI bit Errors on input
ccErrors	optional int Number of CC errors on the TS
services	list of ServiceSourceInfo Info on the services in TS
dejitterStatus	optional DejitterStatus Dejitter Status on Input
path1	optional SrtInputPathStatus Status for single input, or status for input path 1 if input is configured as seamless.
path2	optional SrtInputPathStatus Status for input path 2 if input is configured as seamless.
seamlessStatus	optional SeamlessStatus Seamless status for input if configured.

32.3.49 SrtOutputConnectionStatus

SRT Output Connection Status

struct

socketStatus	SrtSocketState Current SRT socket status
connectionStatus	optional SrtConnectionStatus SRT connection status, if connected
totalPackets	bigint Total number of packets, including retransmissions
totalRetransmittedPackets	bigint Total number of retransmitted packets

32.3.50 SrtOutputConnectionStatusPair

SRT Output Status pair

struct

path1	optional SrtOutputConnectionStatus Status for single output, or status for output path 1 if output is configured as cloned.
-------	---

path2

optional SrtOutputConnectionStatus

Status for output path 2 if output is configured as cloned.

32.3.51 SrtOutputStatus

SRT Output Status

struct

status

list of **SrtOutputConnectionStatusPair**

List of status per active stream

activeConnections

int

Number of active SRT connections

connectionCountTotal

int

Accumulated number of successful SRT connections

32.3.52 SrtSocketState

SRT Socket State

enum

CLOSED

SRT socket is not connected

CONNECTED

SRT socket is connected

32.3.53 SrtStatusInputRes**struct**

data

map from **UUID** to **SrtInputStatus****32.3.54 SrtStatusOutputRes****struct**

data

map from **UUID** to **SrtOutputStatus****32.3.55 SrtStatusReq****variant**

ids

list of **UUID****32.3.56 TimeSource**

Time Source

enum

PTP

PTP (Precision Time Protocol) time source

NTP

NTP (Network Time Protocol) time source

NoSource

No time source

32.3.57 ZixiConnectionStatus

Zixi Connection State

enum

ZIXI_DISCONNECTED	Connection is closed
ZIXI_CONNECTING	Connection is being established
ZIXI_CONNECTED	Connection is established
ZIXI_DISCONNECTING	Connection is being closed
ZIXI_RECONNECTING	Connection is being established after connection loss

32.3.58 ZixiInputPathStatus

Zixi Input Path Status Status for input if the input is single, or one path if input is seamless

struct

zixiStatus	optional ZixiStatus Zixi specific status for path
inbandLinkStatus	optional InbandLinkStatus Inband Link Status for path

32.3.59 ZixiInputStatus

Input Status for Zixi Inputs

struct

hasBitrate	bool True if input TS has bitrate
rtpErrors	optional int RTP Errors on input
syncByteError	optional int Sync Byte Errors on input
teiBitError	optional int TEI bit Errors on input
ccErrors	optional int Number of CC errors on the TS
services	list of ServiceSourceInfo Info on the services in TS
dejitterStatus	optional DejitterStatus Dejitter Status on Input
path1	optional ZixiInputPathStatus Status for single input, or status for input path 1 if input is configured as seamless.
path2	optional ZixiInputPathStatus Status for input path 2 if input is configured as seamless.
	optional SeamlessStatus

seamlessStatus

Seamless status for input if configured.

32.3.60 ZixiInputStatusReq

Request and response to deliver ZixiInputStatus

variant

ids

list of UUID

32.3.61 ZixiInputStatusRes**struct**

data

map from UUID to ZixiInputStatus

32.3.62 ZixiOutputStatus

Zixi Output Status

struct

path1

optional ZixiStatus

Status for single output, or status for output path 1 if output is configured as cloned.

path2

optional ZixiStatus

Status for output path 2 if output is configured as cloned.

32.3.63 ZixiOutputStatusReq

Request and response to deliver ZixiOutputStatus

variant

ids

list of UUID

32.3.64 ZixiOutputStatusRes**struct**

data

map from UUID to ZixiOutputStatus

32.3.65 ZixiStatus

Zixi Specific Input Status

struct

connection

ZixiConnectionStatus

Status for connection to Zixi broadcaster

upTime

bigint

Lifetime of current connection, given in seconds

rtt	int Round trip time given in milliseconds
jitter	int jitter time in milliseconds
totalPackets	bigint Total number of packets delivered
droppedPackets	bigint Number of packets dropped
notRecoveredPackets	bigint Number of nonrecovered packets
fecRecoveredPackets	bigint Number of recovered packets by forward error correction
arqRecoveredPackets	bigint Number of recovered packets by automatic repeat request

33 streamtype (1.0)

33.1 Type Reference

33.1.1 StreamType

StreamType as defined in ISO 13818-1 Table 2-29.

enum

MPEG_VIDEO	
MPEG_AUDIO	
TTX	
DVBSUB	
PRIVATE	
AC_3	
H264	
VBI	
AAC	
VC_1	
EAC3	
AIT	
HBBTV_CAROUSEL	
DATA_CAROUSEL	
H265	
S302M	
ATSC_MPEG2_VIDEO	
ATSC_AC3_AUDIO	
ATSC_SCTE_35	
ATSC_ENHANCED_AC3_AUDIO	

34 triggers (1.0)

34.1 Command Reference

34.1.1 GetTriggers

Retrieve current Triggers object from database.

- message **GetTriggers.Request**
- message **GetTriggers.Response**

34.1.2 SetTriggers

Update Triggers object in database.

- message **SetTriggers.Request**
- message **SetTriggers.Response**

34.1.3 GetConfigurableTriggers

Returns a map of all configurable triggers. The key is the alarm ID and the value is the human-readable name of the trigger override.

- message **GetConfigurableTriggers.Request**
- message **GetConfigurableTriggers.Response**

34.2 Type Reference

34.2.1 AlarmId

A string uniquely identifying an alarm in the system. See Alarm Documentation for details. Should contain both namespace and ID (e.g., "stream_analysis/pmt_missing").

string

34.2.2 ConfigurableTriggerAlarmIds

Set of alarm IDs for alarms that supports trigger overriding.

set of **AlarmId**

34.2.3 GetConfigurableTriggers.Request

empty **struct**

34.2.4 GetConfigurableTriggers.Response

struct

configurableTriggers

map from **AlarmId** to **TriggerName**

34.2.5 GetTriggers.Request

empty **struct**

34.2.6 GetTriggers.Response

Triggers

34.2.7 SetTriggers.Request

Triggers

34.2.8 SetTriggers.Response

empty **struct**

34.2.9 TriggerActive

Boolean specifying the state of a trigger override.

bool

34.2.10 TriggerConfig

Each trigger corresponds to an alarm, but trigger overrides do not impact the alarms themselves. If a trigger is disabled, the alarm will appear but the flow will not be considered broken.

struct

triggers **map** from **AlarmId** to **TriggerActive**

34.2.11 TriggerName

Human-readable name used to display a trigger override in the UI.

string

34.2.12 Triggers

- Object stored in the database containing the Trigger Configurations
- The config contains a map of all trigger overrides that should apply globally for all flows on this card.
- The map should contain all the triggers returned by GetConfigurableTriggers.
- For all configurable triggers not present, a default TriggerActive will be used instead.
- Default TriggerActive is True for Critical alarms and False for alarms with other severities.
-

struct

config **TriggerConfig**

34.2.13 ValidateTriggerConfigResult

struct

validTriggers	set of AlarmId
invalidTriggers	set of AlarmId
missingTriggers	set of AlarmId

35 tsstatus (1.5)

35.1 Overview

Changelog:

1.5

- Added
 - RPC `GetOutputPidStatus` was added, which takes in a list of uuid of outputs and returns the status of the respective output transport streams.

1.4

- Changed
 - component from 1.4 to 1.5

1.3

- Changed
 - Dependency module component bumped from version 1.1 to 1.4

35.2 Command Reference

35.2.1 `GetOutputPidStatus`

Get output PID status

- message `GetOutputPidStatus.Request`
- message `GetOutputPidStatus.Response`

35.3 Type Reference

35.3.1 `GetOutputPidStatus.Request`

`OutputPidStatusRequest`

35.3.2 `GetOutputPidStatus.Response`

`OutputPidStatusResponse`

35.3.3 `OutputPidStatusRequest`

Output PID status request

`struct`

outputs `list of UUID`

35.3.4 `OutputPidStatusResponse`

Response of an output PID status request

`struct`

data

map from **UUID** to **OutputTransportStreamStatus**

35.3.5 OutputTransportStreamStatus

Status for the output TS

struct

services

map from **int** to **ServiceSourceStatus**

Status for each service in the TS. Map from service ID to status struct.

pids

list of **PidStatus**

Info on all PIDs, both signalled and unsignalled. Map from service PID value to status struct.

35.3.6 PidStatus

PidStatus state of PID

struct

serviceId

list of **int**

List of serviceIds a PID belong to.

componentType

optional **component.ComponentType**

ComponentType of PID

ccErrors

int

CC errors on PID

dataRate

bigint

Data rate of PID

scrambling

Scrambling

Scrambling of PID (none, even or odd)

PCR

bool

present

bool

True if signalled pid has rate

language

optional **string**

Language of PID, if applicable (e.g. audio pid)

casId

optional **int**

Id of CAS system, if applicable (e.g. ECM pid)

pidIn

int

PID number on input

pidOut

optional **int**

PID number on output

35.3.7 Scrambling

Scrambling state of PID

enum

NONE

No scrambling

EVEN

Scrambling with even control word

ODD

Scrambling with odd control word

35.3.8 ServiceSourceStatus

Status for an input service

struct

serviceName	optional string Service name
pmtPid	int PMT pid for service
pcrPid	optional int PCR pid for service if present

35.3.9 TransportStreamStatus

Status for the TS

struct

ccErrors	int Total CC error count for this TS.
services	map from int to ServiceSourceStatus Status for each service in the TS. Map from service ID to status struct.
pids	map from int to PidStatus Info on all PIDs, both signalled and unsignalled. Map from service PID value to status struct.
tsId	optional int Transport Stream ID; set if PAT pid present in stream
originalNetworkId	optional int Original Network ID; set if SDT pid present in stream

36 zixi (1.1)

36.1 Type Reference

36.1.1 ZixiAlgorithm

Zixi Encryption Algorithm

enum

AES_128	AES with 128-bit key
AES_192	AES with 192-bit key
AES_256	AES with 256-bit key
CHACHA20	Chacha20 with 256-bit key

36.1.2 ZixiClonedTransport

Zixi Cloned Transport

struct

pathA	ZixiTransportSettings Cloned Path A
pathB	ZixiTransportSettings Cloned Path B

36.1.3 ZixiEncryption

Zixi Encryption

struct

algorithm	ZixiAlgorithm Encryption algorithm
key	string Encryption key as a hex string

36.1.4 ZixiInputSettings

Zixi Input Settings

struct

transport	ZixiInputTransport
decryption	optional ZixiEncryption
maxLatency	int

36.1.5 ZixiInputTransport

Zixi Input Transport

Seamless or single input

variant

seamless	ZixiSeamlessTransport Seamless input
single	ZixiTransportSettings Single input

36.1.6 ZixiOutputSettings

Zixi Output Settings

struct

transport	ZixiOutputTransport
maxLatency	int
encryption	optional ZixiEncryption
tsPacketsPerFrame	int
maxBitrate	int

36.1.7 ZixiOutputTransport

Zixi Output Transport

Cloned or single output

variant

cloned	ZixiClonedTransport Cloned output
single	ZixiTransportSettings Single output

36.1.8 ZixiSeamlessTransport

Zixi Seamless Transport

struct

pathA	ZixiTransportSettings Seamless Path A
pathB	ZixiTransportSettings Seamless Path B
preferred	ZixiSeamlessTransport.preferred Preferred pipe
seamlessBufferSize	int Input buffer size. Buffer for reconstructing missing packets on the input. Ranges 1-400ms

36.1.9 ZixiSeamlessTransport.preferred**enum**

A
B
FLOATING

36.1.10 ZixiTransportSettings

Zixi Transport Settings

struct

interfaceId	UUID UUID of the network interface
streamId	string Stream id, as indicated by the broadcaster
streamPassword	optional string Stream password, optional field used for authentication
remoteHost	string The IP-address of the broadcaster
remotePort	int Port of the remote broadcaster in the range [1024-65535]
localPort	int Local UDP port for Zixi traffic

37 zixiValidate (1.0)

37.1 Type Reference

37.1.1 IntRange

Integer Range

struct

min	int Minimum value of the range.
max	int Maximum value of the range.

37.1.2 Minimum

Minimum

struct

min	int Minimum value of the range
-----	--

37.1.3 Result

Validation result

struct

result	bool True if config is valid and false otherwise.
msg	optional string An error message, which may be given if the result is invalid.